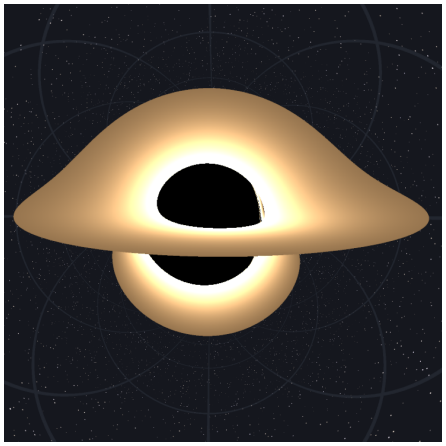
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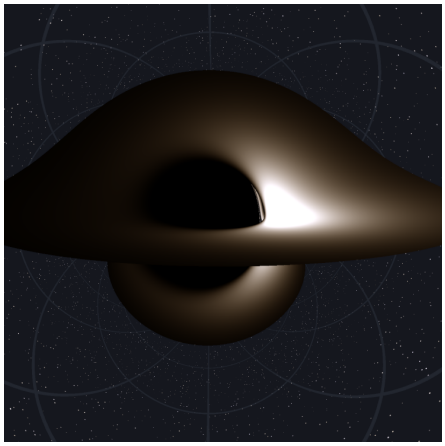
A First-Principles Relativistic Simulator of the Galactic-Centre Black Hole

Shadow metrology, S-star orbit fitting, and a free-quadrupole test of the no-hair theorem — with a complete account of the methods, tests, and development of the instrument

Kerr $a_* = 0.94$, band disk



Novikov–Thorne + beaming



C. Excelsis

Independent research report

July 6, 2026

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Abstract

We present a ground-up, physically rigorous simulator of the Galactic-centre black hole Sgr A*, built on exact Kerr geodesics and validated end-to-end against published strong-field observations. Null and timelike geodesics are integrated in a second-order Mino-time formulation of the Carter equations with per-step constraint projection, conserving (E, L_z, Q) to better than 10^{-6} relative over complete orbits. A CPU reference ray tracer reproduces the analytic Schwarzschild shadow to a relative accuracy of 5×10^{-16} and the Bardeen critical curve for spinning holes; a double-precision GPU compute port agrees with the CPU ground truth to a mean pixel residual of 0.00289. Fitting 13 orbital and potential parameters to the 145 astrometric and 44 radial-velocity measurements of the star S2 (Gillessen et al., 2017) with full general-relativistic timelike geodesics recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8272 \pm 182$ pc, within 0.14σ and 0.03σ of the GRAVITY interferometric values, at a reduced χ^2 of 0.959. The model reproduces the gravitational-redshift excess at the 2018 pericentre passage ($205\,\mathrm{km\,s^{-1}}$ peak) detected by GRAVITY Collaboration (2018), and the Schwarzschild precession of 11.98 arcmin per orbit confirmed by GRAVITY Collaboration (2020). At the GRAVITY mass and distance the simulated photon ring diameter is $49.68\,\mu$as, inside the EHT measurement of $51.8 \pm 2.3\,\mu$as (Event Horizon Telescope Collaboration, 2022). Freeing the quadrupole moment in a Hartle–Thorne extension yields $\varepsilon_q = 0.00 \pm 0.93 \times 10^5$ (68 per cent), consistent with the no-hair theorem.

This report is the long-form account of that programme. Part I is a plain-language primer for readers without a physics background. Parts II–IV derive, from first principles, every formula the simulator implements; document the numerical methods and their error analysis; present the full results with their statistical treatment; and record the complete development history, test suite, and validation gates through which the instrument was built.

Notation and Conventions

Geometric units $G = c = 1$ are used throughout unless stated; masses, lengths and times are quoted in units of the gravitational radius M (for Sgr A*: $GM/c^2 = 5.1243^{-1}$ scaling on the sky, $GM/c^3 = 21.17$ s). Metric signature $(-,+,+,+)$; Boyer–Lindquist coordinates (t, r, θ, ϕ).

Reproducibility Statement

Every quantitative statement in this report that is not a cited literature value is typeset from a macro in `numbers.tex`, generated by `scripts/make_numbers.py` from machine-readable artefacts produced by the simulation; every figure is generated by `make figures` (matplotlib) from the same artefacts. The single command `make reproduce` re-runs the full chain — unit tests, reference renders, sub-ring metrology, orbit fits, calibration, sweeps, and figure generation — from source. A number that cannot survive this cycle cannot appear in this document.

Part I

Foundations for the Non-Specialist

Chapter 1

Gravity, Mass, and the Idea of a Black Hole

This first part of the report assumes no physics background at all. It explains, in ordinary language, every idea the technical chapters build on. If you are comfortable with general relativity, skip directly to Part II.

1.1 Throwing things upward

Throw a ball into the air and it comes back. Throw it harder and it climbs higher before it comes back. This everyday fact hides a precise question: *is there a throwing speed so great that the ball never comes back at all?*

There is. For every planet or star there is a special speed, the *escape velocity*, at which a thrown object just barely coasts away forever, slowing all the while but never quite stopping. From the surface of the Earth it is about 11 kilometres per second — thirty times the speed of sound. From the surface of the Sun it is about 620 kilometres per second. The rule of thumb is simple: the more mass squeezed below you, and the closer you stand to it, the faster you must go to leave.

That second part — *the closer you stand* — is the key to everything in this report. Squeeze the same mass into a smaller ball and its surface moves closer to the centre, so the escape velocity from that surface goes up, even though the total mass never changed. Squeeze the Sun to a radius of three kilometres and the escape velocity at its surface would reach the fastest speed there is: the speed of light, about 300 000 kilometres per second.

1.2 The point of no return

Light's speed is not just fast; physics as we understand it (and as Chapter 2 sketches) says nothing can move faster. So an object compressed within that critical size has a startling property: at some radius, not even light travels fast enough to escape. Whatever falls inside can send no signal out. The sphere where escape would require exactly the speed of light is called the *event horizon*: an "event" happening inside it can never be seen from outside, even in principle.

An object like this is a *black hole* — black because no light leaves it, and a hole because things can fall in but not climb out. The critical radius is named the *Schwarzschild radius* after Karl Schwarzschild, who solved Einstein's equations for a single round mass within weeks of their publication in 1915. The Schwarzschild radius grows in direct proportion to the mass: double the mass and the horizon doubles in size. For the Earth it is about nine millimetres — the Earth would have to be crushed to the size of a marble. For the Sun, about three

kilometres. For the black hole at the centre of our Galaxy, which weighs about 4.3 million Suns, it is around 13 million kilometres — some eighteen times the radius of the Sun, yet still tiny on galactic scales.

Two cautions about the picture this conjures. First, the horizon is not a surface you could touch: nothing is *there*. It is a boundary drawn in empty space, the line on the map beyond which no path leads back out. A falling astronaut would notice nothing special while crossing it. Second, the mass is not spread throughout the dark sphere. As far as the theory can say, everything that fell in is concentrated at the centre in a region where the equations lose their meaning — the *singularity*. The singularity is a statement of ignorance rather than a physical prediction: it marks where the current theory of gravity stops being trustworthy. Fortunately, for everything in this report only the *outside* of the horizon matters, and there the theory is precise and testable.

1.3 Sagittarius A*, our own giant

Point a telescope at the constellation Sagittarius and you look toward the centre of the Milky Way, 8 277 parsecs away (a *parsec* is about 3.26 light-years; the light arriving tonight left before the last ice age peaked). At the exact dynamical centre sits a compact radio source named Sgr A* ("Sagittarius A star"). Around it, stars swarm at extraordinary speeds — a few pass their closest approach at several thousand kilometres *per second*, about a hundred times faster than the Earth orbits the Sun. Something enormously heavy and enormously compact must be holding them: the measured mass is 4.297 ± 0.012 million solar masses, packed into a region smaller than the orbit of Mercury. No cluster of dim stars, no exotic gas cloud survives scrutiny as an alternative; a supermassive black hole is the only explanation standing (Chapter 7 gives the observational history).

Sgr A* is the star of this report for a practical reason: it is the *nearest* supermassive black hole, and the only one where two entirely different kinds of measurement meet. We can follow individual stars looping around it year by year (Chapters 5, 14, 15), and we can photograph the glowing ring of light bent around its horizon (Chapters 3, 11). This report describes a simulator built from first principles that reproduces both kinds of measurement from the same underlying equations.

1.4 A sense of scale

The numbers in this subject span so many powers of ten that it is worth pausing to place them on one line (Fig. 1.1).

Astronomers measure distances within a planetary system in *astronomical units* (au): one au is the Earth–Sun distance, about 150 million kilometres. The horizon of Sgr A* spans roughly a quarter of an au. The star S2, the most important character of Part III, dives to about 120 au from the black hole — about three times the distance of Pluto from the Sun — which sounds close until you recall that this is still about 1 500 horizon radii out.

For things on the sky, astronomers use angles. A degree is what your little finger covers at arm's length; an *arcsecond* is 1/3600 of a degree, roughly a coin seen from four kilometres away; and the measurements in this report are made in *milliarcseconds* (a thousandth of an arcsecond) and *microarcseconds* (μas, a millionth). The photon ring of Sgr A* spans about 52 μas — the apparent size of a doughnut on the Moon. That such an angle is measurable at all, by linking radio telescopes across the whole Earth into one instrument, is among the technical wonders of modern astronomy.

One more unit will keep appearing: the *gravitational radius*, written M in this report. It is the natural yardstick of a black hole — half the Schwarzschild radius — and it lets us describe

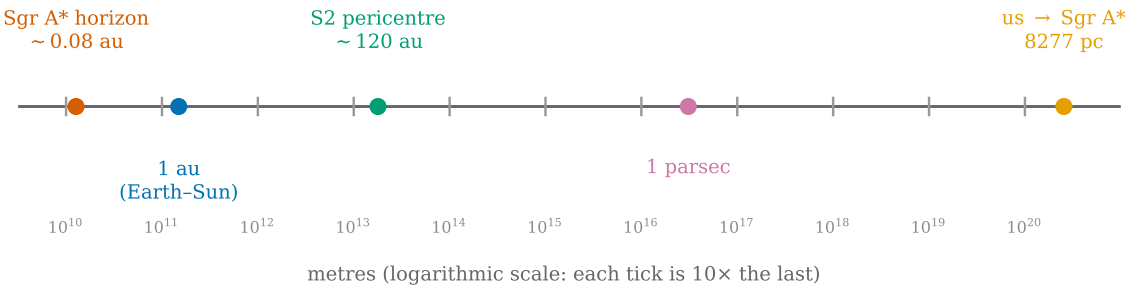


Figure 1.1: The scales of the problem on a single logarithmic line, from the Sgr A* horizon to its distance from us. Each tick is ten times the previous one: the distance to Sgr A* is about twenty billion times the size of its horizon — comparable to viewing a coin at the distance of the Moon.

any black hole, whatever its mass, with the same numbers. "The disk's inner edge sits at $6\,M$" is true of a stellar black hole and of Sgr A* alike; only the metre-stick changes. Chapter 6 says more about this trick of working in natural units.

Terms you just met.

Escape velocity The launch speed at which an object barely coasts away from a mass forever.

Event horizon The boundary around a black hole from inside which nothing, not even light, can escape. Not a material surface.

Schwarzschild radius The size of the event horizon of a non-spinning mass; proportional to the mass.

Singularity The central region where the known equations of gravity break down; hidden inside the horizon.

Sgr A* The 4.3-million-solar-mass black hole at the centre of the Milky Way, 8 277 parsecs away.

au, parsec, milli-/microarcsecond Units for distances and sky angles; see Fig. 1.1.

Gravitational radius (M) The natural length unit of a black hole; the horizon of a non-spinning hole sits at $2\,M$.

→ *Full treatment: the observational case for Sgr A* in Chapter 7; the exact horizon formulas in Chapter 8.*

Chapter 2

Spacetime and Einstein's Gravity, Gently

2.1 Gravity is not a pull

Newton described gravity as a force: the Earth pulls the apple, the Sun pulls the Earth. His description is astonishingly accurate for slow objects far from big masses, and Chapter 5 will use it freely. But it leaves a mystery — how does the Sun reach across empty space to tug on the Earth? — and it fails, measurably, exactly in the situations this report cares about.

Einstein's 1915 answer, *general relativity*, removes the mystery by removing the force. In his picture, mass does not pull on distant objects. Instead, mass changes the shape of *spacetime* — the four-dimensional fabric of the three space directions plus time — in its neighbourhood, and objects then simply travel through that reshaped fabric in the straightest possible lines. The famous slogan, due to John Wheeler, is: *matter tells spacetime how to curve, and spacetime tells matter how to move.*

A useful (if imperfect) picture is a stretched rubber sheet with a bowling ball resting in the middle. Marbles rolled across the sheet curve toward the ball, not because the ball grips them, but because the surface they travel on slopes. Each marble is doing the most natural thing possible — going straight ahead on a surface that is not flat. The Earth orbits the Sun for the same reason: it moves straight ahead through a region of spacetime that the Sun has curved, and "straight ahead" there is a loop.

The straightest possible path through curved spacetime has a name that appears on nearly every page of Part II: a *geodesic*. On the Earth's curved surface, geodesics are the great circles that airliners follow; in curved spacetime, geodesics are the paths of everything that falls freely — thrown stones, orbiting planets, and light itself. The entire simulator described in this report is, at its heart, a machine for computing geodesics very accurately.

2.2 Light bends

Because light also travels through spacetime, its path also bends where spacetime curves. The effect is tiny near the Sun — starlight grazing the solar surface deflects by less than two arcseconds, first measured during the 1919 eclipse — but near a black hole the bending becomes wild.

Fig. 2.1 shows real computed light paths around a non-spinning black hole, calculated with the same equations the simulator uses. A ray passing far away bends gently, like the starlight of 1919. A ray passing closer bends more. And there is a critical aiming distance at which the bending wins completely: the ray wraps around the hole and can circle it — light in orbit. Aim any closer and the ray spirals in and is gone. The dividing line between "escapes, however

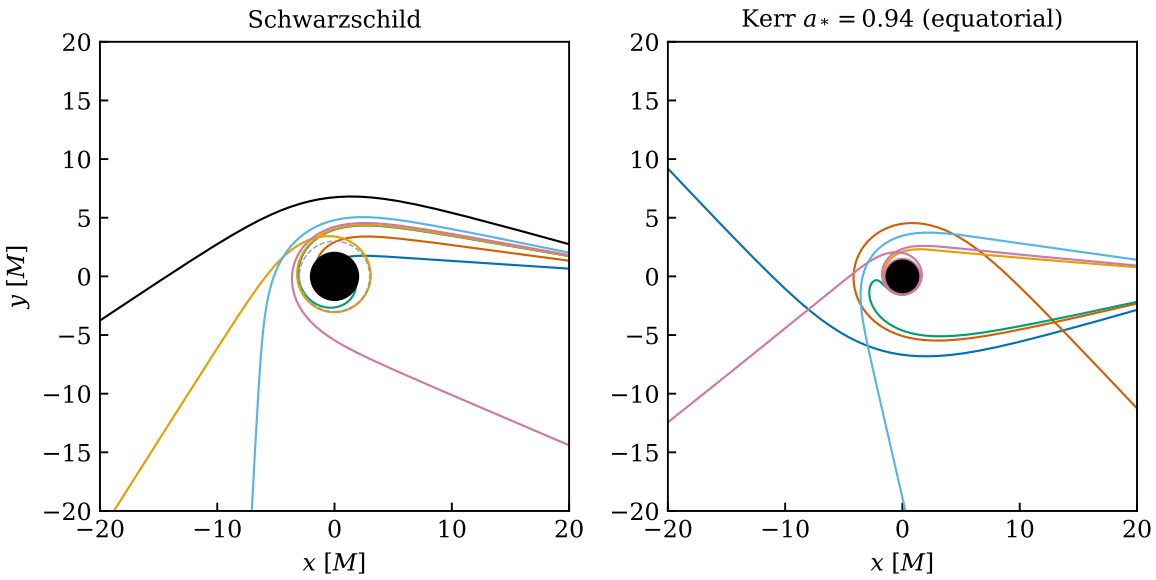


Figure 2.1: Computed paths of light rays in the equatorial plane of a black hole, from the simulator's own geodesic integrator. Left: a non-spinning (Schwarzschild) hole — rays aimed farther out escape with a bend; a critical ray circles on the dashed photon sphere; closer rays are captured. Right: a rapidly spinning (Kerr) hole — the spacetime itself circulates, so rays travelling with the spin behave differently from rays travelling against it (Chapter 3).

bent" and "captured" is what carves out the black-hole *shadow*, the subject of Chapter 3.

This bending also acts as a distorting lens. A black hole with a glowing disk around it shows you the disk twice over: light from the part of the disk hidden *behind* the hole is bent up and over the top (and under the bottom), so the far side appears above and below the dark centre. That is why the famous images — the Event Horizon Telescope's photographs and the film *Interstellar*'s Gargantua — have their characteristic crossed-halo look. Every one of those features appears, for the right physical reasons, in the renders of Part III.

2.3 Time runs slow near mass

Curved spacetime does something with no Newtonian counterpart at all: it changes the rate of time. Clocks deeper in a gravitational well genuinely run slower as seen from outside — not because of any mechanical fault, but because "one second" is itself a local quantity, like "north", that depends on where you are.

The effect is real and routine: the satellites of the GPS system carry clocks about 20 000 km above the Earth, where time runs faster than at the ground by about 45 microseconds per day. Uncorrected, satellite navigation would drift by kilometres within a day. Near a black hole the same effect grows enormous; at the horizon, an outside observer would see a falling clock slow to a stop.

For this report the significance is very concrete: when the star S2 dives to its closest approach to Sgr A*, its light is emitted from slightly deeper in the black hole's gravitational well, where time runs slower. Every wave of its light is stretched on the climb out — the wavelength shifts toward the red. The GRAVITY instrument measured exactly this *gravitational redshift* in 2018, and reproducing that measurement from first principles is one of this report's central results (Chapters 4, 15).

2.4 What a metric is (without tensors)

One piece of vocabulary unlocks all of Part II. To compute anything, we need coordinates — labels like latitude and longitude that name the points of spacetime. But coordinates are just names; they do not tell you how *far apart* two nearby points really are. On the Earth, one degree of longitude is 111 km at the equator and zero at the pole: the conversion from coordinate differences to true distances varies from place to place.

The rulebook for that conversion is called the *metric*. Give the metric the small coordinate steps between two nearby events (a bit of north–south, a bit of east–west, a bit of time) and it returns the true separation — including the true amount of elapsed clock time. All of the curvature of spacetime is encoded in how this rulebook changes from place to place; slow-running clocks and bending light are both just entries in the rulebook. Einstein's field equations determine the metric produced by a given arrangement of mass; for a lone spinning black hole the exact answer is known and is the famous *Kerr metric* (Chapters 3, 8), on which this entire simulator is founded.

When Part II writes symbols such as g_{tt} or g_{rr}, nothing deeper is meant than "the entry of the rulebook that converts time-steps (or radius-steps) into true separations, at this location".

Terms you just met.

Spacetime Space and time treated as one four-dimensional fabric, whose shape is what we experience as gravity.

General relativity (GR) Einstein's theory: mass curves spacetime; freely falling objects follow the straightest paths in it.

Geodesic The straightest possible path through curved spacetime; the track of anything falling freely, including light.

Gravitational lensing The bending of light by curved spacetime; a black hole shows the far side of its own disk, bent over the top.

Time dilation Clocks deeper in a gravitational well run slower; measurable with GPS satellites and with the star S2.

Gravitational redshift Light climbing out of a gravitational well is stretched to longer, redder wavelengths.

Metric The location-dependent rulebook that converts coordinate differences into true distances and times; in GR it carries all of gravity.

→ *Full treatment: the Kerr metric in Chapter 8; geodesics and their conserved quantities in Chapter 9; the traced light-bending figures in Chapters 9 and 11.*

Chapter 3

Spinning Black Holes and the Shadow

3.1 Black holes spin

Almost everything in the universe rotates — planets, stars, whole galaxies — and when a rotating object collapses, its spin is not lost. It concentrates, the way a figure skater speeds up when she pulls her arms in. Black holes born from collapsing, swirling matter are therefore expected to spin, often ferociously.

A non-spinning black hole is described exactly by Schwarzschild's 1915 solution of Chapter 1. The spinning case resisted for another half-century, until Roy Kerr found the exact solution in 1963. The *Kerr metric* — the rulebook of Chapter 2 for a spinning hole — is the mathematical foundation of this entire report, and it says that a black hole is completely described by just two numbers: its mass and its spin. The spin is usually quoted as a dimensionless number a_* between 0 (not spinning) and 1 (the theoretical maximum, spinning so fast the horizon would tear open). For Sgr A* the working estimate used throughout this report is $a_* = 0.94$ — a rapid spinner.

3.2 Frame dragging: spacetime as a whirlpool

A spinning mass does something Newton never imagined: it drags spacetime around with it. Close to a spinning black hole, "standing still" is simply not on the menu — space itself circulates like water around a drain, and everything in it, light included, is carried along. This *frame dragging* has been measured in Earth orbit (by the Gravity Probe B gyroscopes and the LAGEOS satellites) at the level of fractions of a degree per century; near Sgr A* it becomes a dominant feature of the landscape (Fig. 3.1).

There is a region, outside the horizon but close to it, where the dragging is so strong that *nothing* can hold still — to hover without circulating you would need to move faster than light. This region is the *ergosphere*, a flattened shell thickest over the equator (Fig. 3.2). Objects inside it can still escape — it is not a point of no return — but they cannot avoid being swept around.

3.3 The anatomy of a spinning hole

Fig. 3.2 labels the landmarks that Part II computes exactly, for a non-spinning and a rapidly spinning hole side by side.

- The *event horizon* (black): the point of no return. Spin shrinks it, from $2\,M$ at $a_* = 0$ toward $1\,M$ at the extreme.

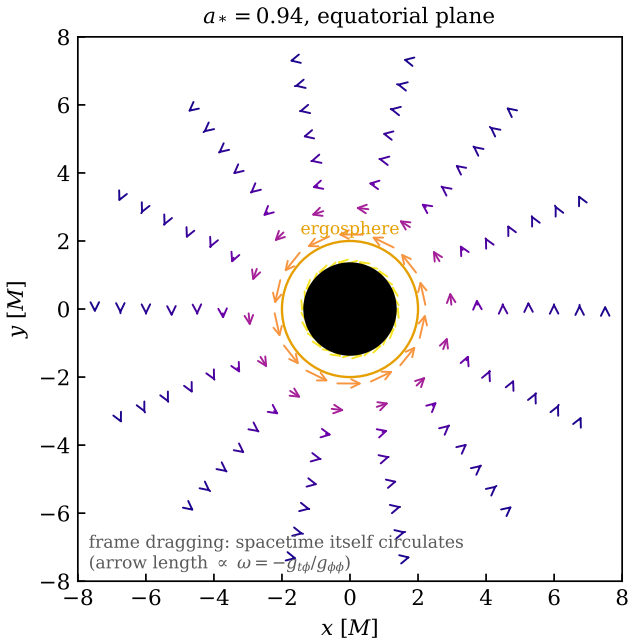


Figure 3.1: Frame dragging around a black hole with $a_* = 0.94$, computed from the Kerr rulebook. Each arrow shows how far a "stationary" marker at that radius is swept around per unit time; inside the orange boundary (the ergosphere) refusing to rotate is impossible.

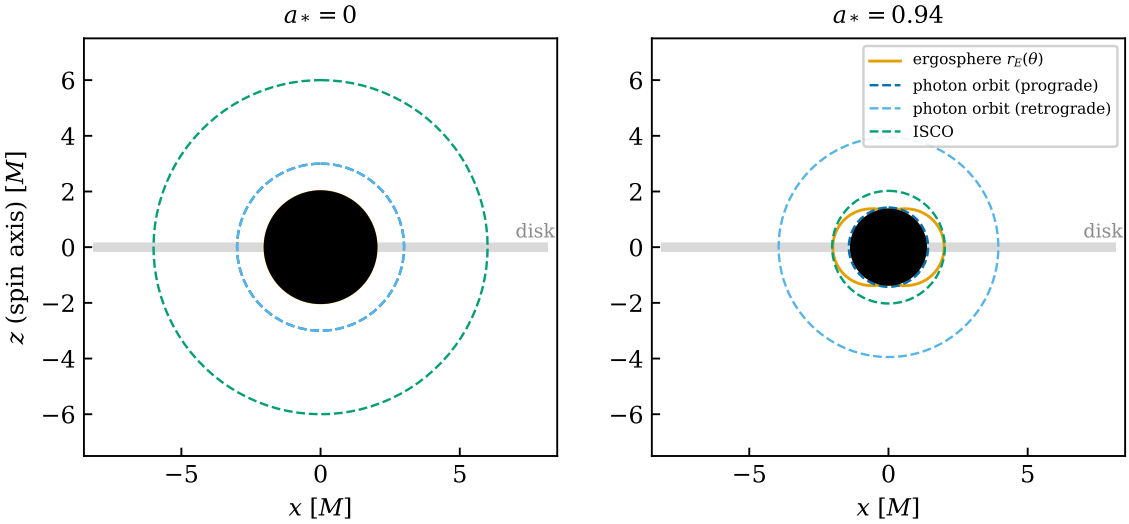


Figure 3.2: Cross-sections through a non-spinning ($a_* = 0$) and a rapidly spinning ($a_* = 0.94$) black hole, drawn to scale from the exact formulas of Chapter 8. Spin shrinks the horizon, opens up an ergosphere, splits the photon orbit into prograde/retrograde pairs, and pulls the innermost stable orbit (ISCO) dramatically inward.

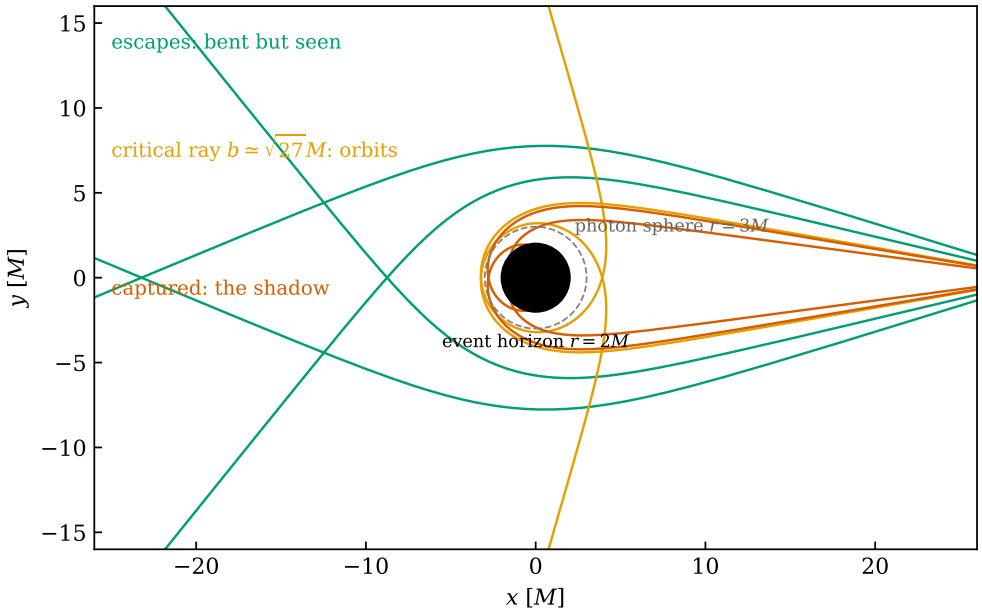


Figure 3.3: How the shadow forms, computed from the exact bending equation. Green rays escape (bent); the orange critical ray circles on the photon sphere; red rays are captured. The shadow is the set of camera directions whose rays end on the horizon — its edge is light that grazed the photon sphere.

- The *ergosphere* (orange): the no-standing-still zone, touching the horizon at the poles and bulging at the equator.
- The *photon orbits* (blue): radii where light can circle the hole. For a spinning hole there are two equatorial ones — a closer orbit for light moving *with* the rotation (*prograde*) and a farther one for light moving *against* it (*retrograde*). Spacetime's whirlpool helps the co-rotating light hold a tighter circle.
- The *ISCO* (green), the innermost stable circular orbit: the closest circle on which matter can orbit steadily. Any closer and the slightest nudge sends it spiralling in. The ISCO is the inner edge of the glowing disk of Chapter 4, and it moves inward spectacularly with spin — from $6\,M$ for $a_* = 0$ to about $2\,M$ at $a_* = 0.94$. Much of what changes visually with spin in our renders traces back to this single fact.

3.4 The shadow and the photon ring

Now surround the hole with light — a glowing disk, or simply the stars behind it — and ask what a distant camera sees. Chapter 2 showed that rays aimed too close are captured. Point the camera at the hole, and there is a patch of sky, for every pixel of which the answer is "that line of sight ends on the horizon". That dark patch is the *shadow* (Fig. 3.3).

Three facts about the shadow do a great deal of work in this report.

First, *the shadow is bigger than the horizon.* Light does not have to hit the horizon to be lost — anything aimed within the critical distance spirals in. For a non-spinning hole the shadow's radius is $\sqrt{27} \approx 5.2\,M$, about 2.6 times the horizon radius. Our simulator reproduces this number to sixteen digits (Chapter 11), which is the sort of check that separates a scientific instrument from a pretty picture.

Second, *the shadow's edge is made of light that almost stayed.* Rays that graze the critical boundary can circle the hole once, twice, three times before escaping to the camera. Each extra half-turn stacks another, exponentially thinner and fainter image of the whole sky (and of the disk) right at the shadow's rim. This onion of nested images is the *photon ring*, and its sub-rings are analysed quantitatively in Chapter 13. The Event Horizon Telescope's measurement of this ring for Sgr A* — $51.8 \pm 2.3\,\mu$as across — is one of the two observational anchors of this whole project.

Third, *spin distorts the shadow.* Frame dragging lets prograde light survive closer in, so the shadow's edge on the side rotating toward you pulls inward: the silhouette flattens on one side and shifts sideways, by an amount our simulator measures precisely (Chapters 11, 19). Viewed nearly pole-on — our actual view of Sgr A* — the distortion is frustratingly small, a few per cent, which is why the ring's *size* alone cannot pin down the spin (the "spin–inclination degeneracy" that Chapter 19 maps out).

Terms you just met.

Kerr metric The exact spacetime of a spinning black hole (Kerr 1963); fully specified by mass and spin.

Spin parameter a_* Dimensionless spin from 0 to 1; Sgr A* is modelled at $a_* = 0.94$.

Frame dragging The whirlpool-like circulation of spacetime around a spinning mass.

Ergosphere The region where standing still is impossible; escape is still possible from inside it.

Prograde / retrograde Moving with / against the black hole's rotation; prograde orbits survive closer in.

Photon sphere / photon orbits Radii at which light circles the hole; unstable, so grazing light eventually escapes or falls.

ISCO Innermost stable circular orbit — the inner edge of the disk; moves from $6\,M$ (no spin) toward the horizon as spin grows.

Shadow The dark patch of captured lines of sight; radius $\sqrt{27}\,M$ for no spin, subtly reshaped by spin.

Photon ring The stack of exponentially fainter images of the scene hugging the shadow's edge.

→ *Full treatment: Kerr geometry and all characteristic radii in Chapter 8; the traced shadow and critical curve in Chapter 11; sub-ring measurements in Chapter 13.*

Chapter 4

Light, Heat, and Colour: Reading the Disk

4.1 Hot things glow, and colour is a thermometer

Heat a poker in a fire and it glows dull red; hotter, and it turns orange, then yellow, then blue-white. This everyday sequence is one of the most useful facts in astrophysics: a dense hot body glows with a spectrum — a recipe of colours — that depends on its temperature and on almost nothing else. Physicists call this ideal glow *blackbody radiation*, and the temperature-to-colour dictionary is exact: around 3 000 degrees looks ember-orange, 6 000 degrees looks white like the Sun's surface, 20 000 degrees looks blue-white. The renders in this report use precisely this dictionary — every pixel's colour comes from a computed temperature pushed through the same colour-matching tables that describe human vision (Chapter 12), not from an artist's palette.

4.2 The accretion disk

Gas falling toward a black hole almost never falls straight in. Like water circling a drain, it has some rotation, so it settles into a flattened, swirling *accretion disk*. Within the disk, gas at each radius orbits at its own rate — inner gas circles faster than outer gas, just as Mercury outpaces Neptune — and the resulting internal friction does two things: it lets the gas drift gradually inward, and it heats the disk white-hot. The energy released is prodigious. Matter spiralling through a disk onto a rapidly spinning black hole can radiate away tens of per cent of its entire rest-mass energy — compared with less than one per cent for nuclear fusion. Our simulator computes this efficiency exactly (it is the η of Chapter 8): about 5.72 per cent for a non-spinning hole, rising to 17.96 per cent at Sgr A*'s spin, because the disk's inner edge (the ISCO of Chapter 3) reaches so much deeper into the well.

The model disk used throughout Part III is the classic *Novikov–Thorne* thin disk: geometrically thin like a vinyl record, hottest just outside the ISCO, cooling outward. It is an idealisation — the real flow around Sgr A* is a hot, puffed-up, far dimmer affair — and Chapter 22 is careful about which conclusions survive that difference. The disk model gives our images their brightness and colour; crucially, the *geometry* of shadow and photon ring comes from the spacetime itself, not from the disk details.

4.3 Why one side of the disk blazes

Every published image of a black-hole disk — the EHT rings, *Interstellar*'s Gargantua, the renders in this report — is conspicuously brighter on one side. That asymmetry is a speed

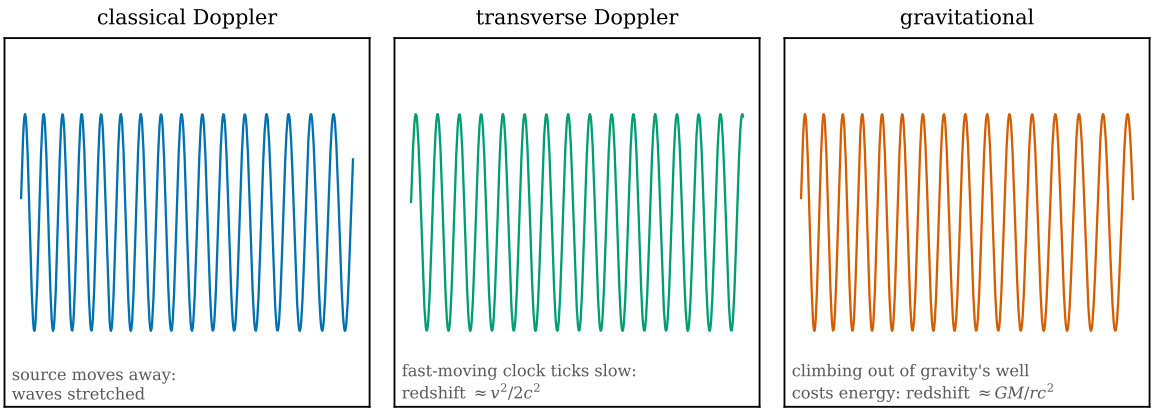


Figure 4.1: The three ways a wavelength gets shifted. Classical Doppler: motion along the line of sight stretches or compresses the wave train. Transverse Doppler: a fast clock ticks slowly, so even sideways motion reddens the light — a purely relativistic effect. Gravitational: light climbing out of a gravity well arrives stretched.

effect, not a lighting effect.

The inner disk orbits at an appreciable fraction of the speed of light. When a source of light moves toward you that fast, three effects conspire: its light is squeezed forward like spray from a moving hose (*beaming*), arrives blueshifted, and arrives at a faster clip. The approaching side of the disk therefore appears far brighter — in our reference render at $a_* = 0.94$, viewed nearly edge-on, the approaching limb outshines the receding one by a factor of 9.60 in total light (Chapter 12). The receding side is correspondingly dimmed and reddened. The whole effect is packaged in a single number per light ray, the *redshift factor* g: the ratio of the light's frequency as received to its frequency as emitted. Brightness scales as g^4 — the fourth power — which is why the asymmetry is so dramatic. If a simulated disk ever looks symmetric, its g^4 is wrong; this was one of our hard validation gates (Chapter 21).

4.4 The three redshifts

That factor g mixes three physically distinct effects, and pulling them apart is exactly what the S2 measurements of Part III require (Fig. 4.1).

1. *Classical Doppler*: the familiar ambulance-siren shift. Motion away from you stretches the waves (redshift); motion toward you compresses them (blueshift). This is by far the largest term for an orbiting star, and Newton would have predicted it too.

2. *Transverse Doppler*: special relativity's addition. A fast mover's clock runs slow, so its emitted waves come out at a reduced rate — a redshift that appears even when the motion is purely sideways. It scales with speed squared, so it only matters when things get truly fast.

3. *Gravitational redshift*: general relativity's addition, from Chapter 2 — light climbing out of a gravitational well is stretched. It scales with depth in the well.

For the star S2 at its 2018 closest approach, the first term is several thousand km/s, while the second and third are each worth only about a hundred km/s — and, by a neat algebraic coincidence worked out in Chapter 15, almost exactly equal to each other. The GRAVITY collaboration's 2018 detection of that little two-hundred-km/s surplus over the

Newtonian prediction was the first direct test of general relativity beside a supermassive black hole. Reproducing it, with no tunable knob, is one of this report's headline results: our model peaks at 205 km/s at epoch 2018.379 (Chapter 15).

Terms you just met.

Blackbody radiation The temperature-determined glow of a dense hot body; the exact temperature-to-colour dictionary used in our renders.

Accretion disk The flattened swirl of hot gas spiralling into a black hole; modelled here as the Novikov–Thorne thin disk.

Accretion efficiency (η) The fraction of rest-mass energy radiated by matter spiralling in: 5.72% (no spin) to 17.96% (Sgr A*'s spin).

Doppler beaming The forward-throwing of light by a fast-moving source; makes the approaching disk limb $\sim$9.60$\times$ brighter in our reference render.

Redshift factor g Received frequency over emitted frequency for one light ray; observed brightness scales as g^4.

Classical / transverse / gravitational redshift The three distinct wavelength shifts; separating them at S2's pericentre is the GRAVITY 2018 test.

$\rightarrow$ Full treatment: disk physics and the g^4 law in Chapter 12; the redshift decomposition and the S2 result in Chapters 14 and 15.

Chapter 5

Orbits, S-Stars, and Weighing a Black Hole

5.1 Kepler's ellipse

Four hundred years before anyone imagined black holes, Johannes Kepler discovered that planets travel on *ellipses* — squashed circles — with the Sun at one *focus*, a point offset from the centre (Fig. 5.1). Three of his terms are used on nearly every page of Part III:

- the *semi-major axis* a — half the long diameter, the orbit's overall size;
- the *eccentricity* e — how squashed and off-centre the ellipse is, from 0 (circle) toward 1 (a stretched sliver);
- the *pericentre* and *apocentre* — the points of closest and farthest approach. (For orbits around stars you may know these as perihelion and aphelion.)

Kepler's second discovery is visible in the dots of Fig. 5.1: bodies move *fast near pericentre and slow near apocentre*. His third links the orbit's size and period to the central mass — bigger orbits take longer, and, decisively for us, *at fixed size, a heavier central mass means a faster orbit*. Measure the size and the period of any orbit and the central mass falls out. This is how we weigh everything in astronomy, from planets to galaxies — and it is exactly how Sgr A* is weighed.

5.2 The S-stars, and S2 above all

Buried in the central light-month of the Galaxy is a swarm of ordinary young stars on extraordinary orbits — the *S-stars*. Since 1992, two teams (in Garching and in Los Angeles) have photographed the region every year, watching individual stars trace out their ellipses around an invisible focus. The star *S2* is the jewel of the collection: bright enough to measure precisely, with a period of just 16 years — short enough that we have now watched it complete more than one and a half full loops. Its orbit is extreme by any planetary standard: eccentricity 0.88, pericentre speed near 7 700 km/s (2.6 % of the speed of light), diving to about 1 500 horizon radii from the black hole.

This report fits the complete published S2 data set — 145 positions and 44 velocities spanning 1992–2016 (Chapter 15) — plus four more S-stars in a joint fit (Chapter 16).

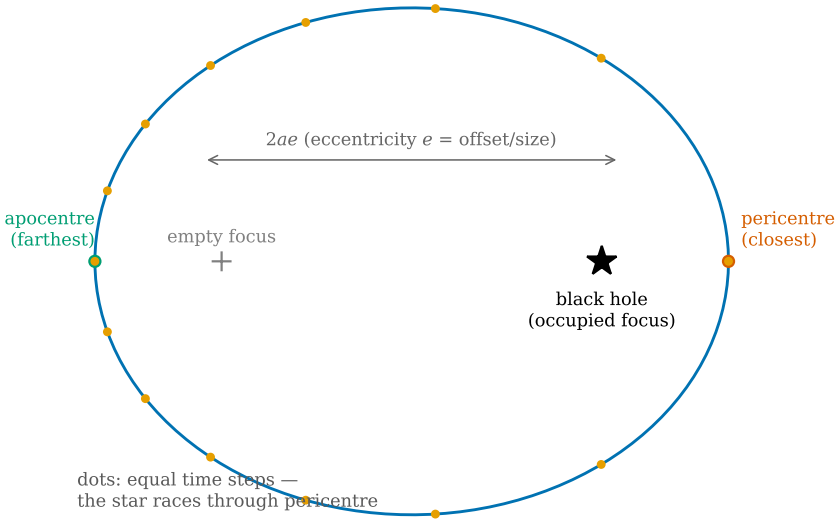


Figure 5.1: The anatomy of a Kepler orbit. The attracting mass sits at one focus; eccentricity measures how off-centre it is. The dots mark equal time steps: the orbiting body loiters near apocentre and races through pericentre.

5.3 Two channels: positions and speeds

Everything we know about the S-stars comes through two complementary kinds of measurement (Fig. 5.2).

Astrometry is simply measuring where the star is on the sky, epoch by epoch — in milliarcseconds, using adaptive-optics cameras on the world's largest telescopes and, since 2017, the GRAVITY interferometer, which links four 8-metre telescopes into one virtual 130-metre one. Astrometry sees the orbit *projected* on the sky, like the shadow of a hoop tilted against a wall.

Spectroscopy measures how fast the star moves *along* the line of sight, via the Doppler shift of a hydrogen absorption line in its spectrum (the Brγ line, in the infrared where the Galactic dust is transparent). This "radial velocity" channel sees exactly the component of motion that astrometry cannot.

Neither channel alone determines everything; together they overdetermine the orbit. And one beautiful consequence deserves its own sentence: because astrometry gives the orbit's *angular* size while the velocities give its *physical* scale, combining them yields the *distance to the Galactic centre* — a geometric distance measurement across 26 000 light-years with no cosmic distance ladder involved. Our fit recovers both the mass and this distance (Chapter 15).

5.4 Einstein's fingerprints on the orbit

If gravity were exactly Newtonian, S2's ellipse would repeat itself forever, fixed in space. General relativity disagrees in two measured ways.

First, the *redshift excess* of Chapter 4: at the 2018 pericentre passage, S2's spectral lines arrived about 200 km/s redder than any Newtonian orbit could explain — time dilation plus gravitational redshift, detected by GRAVITY in 2018.

Second, *orbital precession*: the ellipse itself slowly pivots. Each orbit, the pericentre direction advances by a small angle, so the path traces a rosette rather than a closed ring (Fig. 5.3). For Mercury this famous advance is 43 arcseconds per century and was the first triumph of general relativity; for S2 it is about 12 *arcminutes* per 16-year orbit — some fifty thousand

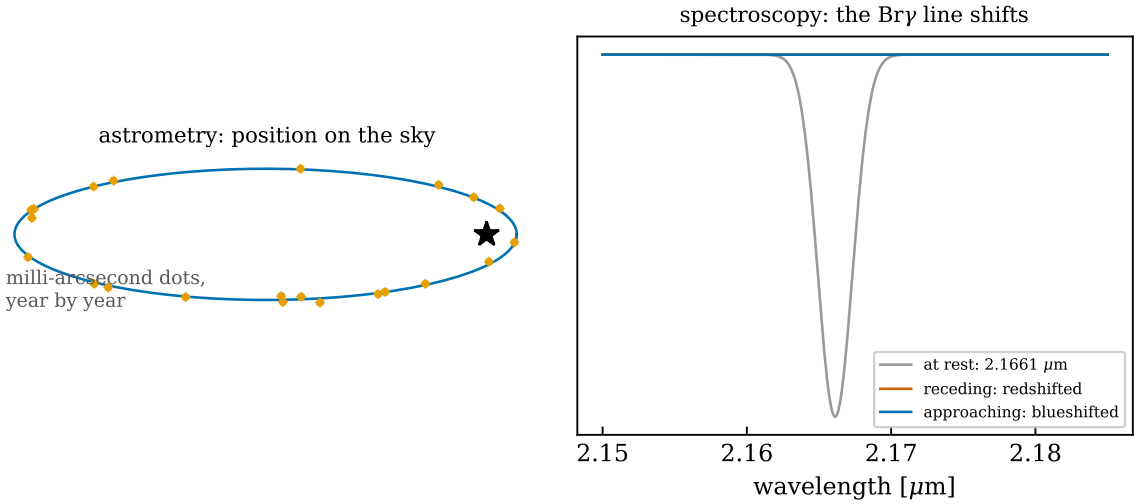


Figure 5.2: The two observation channels. Left — astrometry: the star's position on the sky, measured year after year in milliarcseconds, traces the projected ellipse. Right — spectroscopy: the wavelength of a known line in the star's light shifts with its line-of-sight speed (the Doppler effect of Chapter 4).

times faster — and GRAVITY confirmed it in 2020. Our simulator reproduces both effects with no adjustable amplitudes: the fitted mass and geometry force them (Chapter 15).

5.5 Thirty-four years in one diagram

Fig. 5.4 places the milestones — from the first infrared images of the S-stars to the EHT picture of the ring — on one line, ending with the simulator this report describes: a single instrument that reproduces, from first principles, the measurements of that whole timeline.

Terms you just met.

Ellipse, focus, semi-major axis, eccentricity The shape language of orbits (Fig. 5.1).

Pericentre / apocentre Closest and farthest points of an orbit; S2's pericentre is ∼120 au at 2.6% of light speed.

S-stars, S2 The stars orbiting Sgr A*; S2, with its 16-year period, is the most precisely measured.

Astrometry Measuring sky positions over time (milliarcseconds); gives the projected orbit.

Radial velocity (spectroscopy) Measuring line-of-sight speed from Doppler-shifted spectral lines.

Geometric distance Combining both channels yields the distance to the Galactic centre directly.

Precession / rosette The slow pivot of the orbital ellipse predicted by GR; 12′ per orbit for S2, confirmed 2020.

→ *Full treatment: the relativistic orbit model in Chapter 14; the fit and both GR effects in Chapter 15; the multi-star fit in Chapter 16.*

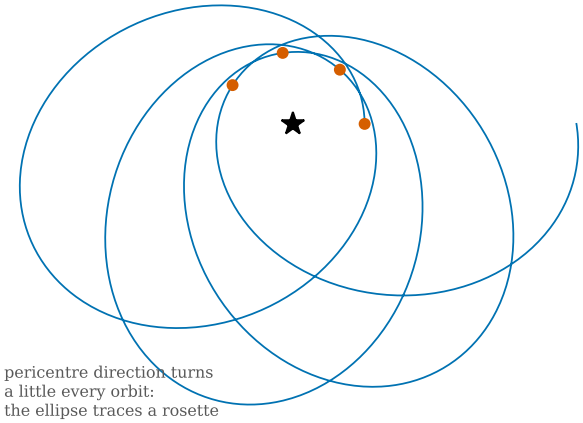


Figure 5.3: Relativistic precession: the orbital ellipse pivots a little each revolution, tracing a rosette. The dots mark successive pericentre directions. (Exaggerated for visibility: S2's advance is $12'$ per orbit; this drawing uses thousands of times more.)

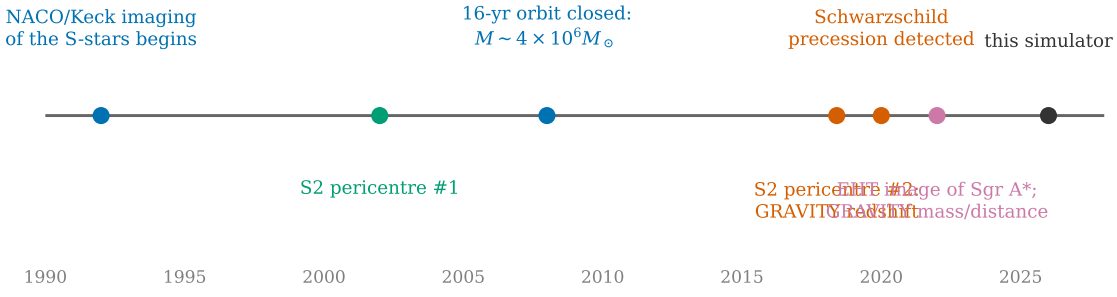


Figure 5.4: Milestones of the Galactic-centre programme covered in this report.

Chapter 6

The Toolkit in Plain Words

This last primer chapter collects the working tools — units, statistics, and one big theorem — that the technical chapters use constantly.

6.1 Natural units: setting $G = c = 1$

Physicists working with black holes almost always set the gravitational constant G and the speed of light c equal to 1. This is not a trick or an approximation; it is a choice of units, exactly like measuring road distances in hours of driving. Once you do it, mass, length, and time all become interchangeable, and a single symbol M can serve as the yardstick for everything: lengths are quoted in units of the gravitational radius ("the ISCO is at $6\,M$"), times in units of the light-crossing time of that radius. For Sgr A*, one M of length is about 0.042 au and one M of time is about 21 seconds. The huge practical benefit is universality: every formula and every rendered image in Parts II–III applies to *any* black hole; only the final conversion to kilometres, seconds, or microarcseconds depends on the actual mass (that conversion is Chapter 19).

A related habit: Part II speaks of *vectors* (arrows with direction and length — a velocity is one) and occasionally *tensors*. For reading this report it is enough to know that a tensor is a machine that eats vectors and returns numbers, in a way that all observers agree on. The metric of Chapter 2 is the only tensor we ever really need: it eats two little coordinate steps and returns their true separation.

6.2 Error bars, and the meaning of χ^2

Every real measurement comes with an uncertainty — an *error bar* — and the whole craft of fitting models to data is bookkeeping those error bars honestly. The universal bookkeeping device is called *chi-squared* (χ^2), and it answers one question: *by how many error bars, in total, does the model miss the data?* (Fig. 6.1). For each data point, take the miss, divide by that point's error bar, square it (so misses in either direction count alike, and big misses count a lot), and add up over all points.

Two rules of thumb make χ^2 readable. A model that is right, fitted to honest error bars, should miss each point by about one error bar on average — so χ^2 divided by the number of data points (the *reduced* χ^2) should be near 1. Much bigger means the model (or the error bars) are wrong; much smaller means the error bars were pessimistic. Our headline S2 fit lands at $\chi^2_\nu = 0.959$ (Chapter 15) with the published error bars used exactly as given.

And once the best knob-settings are found, *how well known* is each knob? Wiggle a knob away from its best value and χ^2 climbs; the range over which it climbs by less than a set amount is the knob's *confidence interval* — the "±" quoted after every fitted number in this report. When we write $M = 4.270 \pm 0.195$ million suns, the $\pm$ is precisely such an interval.

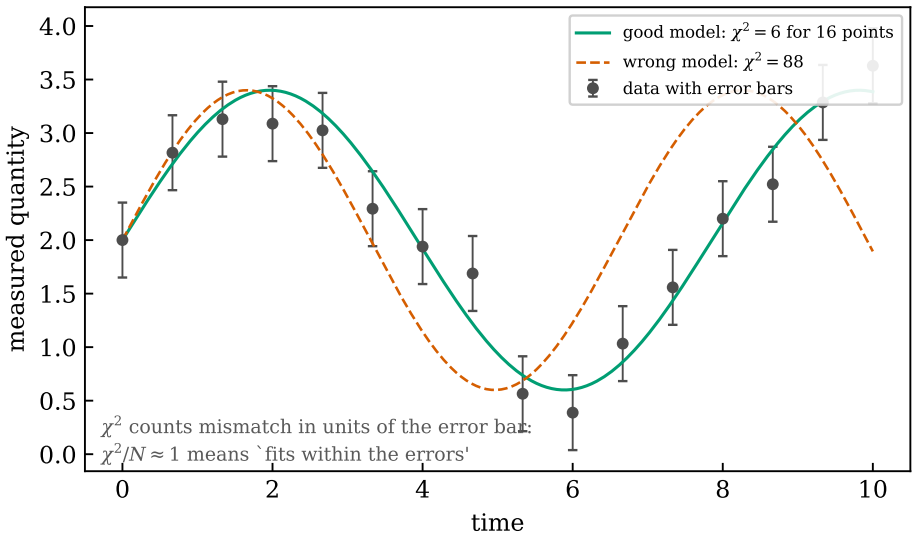


Figure 6.1: χ^2 made visible. Two candidate models through the same data: the good one misses each point by about one error bar ($\chi^2 \approx$ number of points); the wrong one racks up an enormous total. Fitting means turning model knobs until χ^2 is as small as it gets.

6.3 MCMC: mapping uncertainty by wandering

For a model with thirteen knobs, the surface of χ^2 over all knob combinations is a thirteen-dimensional landscape, and simple wiggling can miss how knobs trade off against one another. The heavy machinery for this is *Markov chain Monte Carlo* (MCMC): release a crowd of random walkers into the landscape, with rules that make them drift toward — and then wander within — the low-χ^2 valleys (Fig. 6.2). After enough steps, *where the walkers spend their time is itself the answer*: dense regions are probable parameter values, sparse regions improbable. Histograms of the walkers' positions give every parameter's uncertainty and every pairwise trade-off at once. Chapter 15 runs exactly this (32 walkers, thousands of steps, the full relativistic model evaluated at every step) and checks that the walker map agrees with the simpler wiggle estimate.

6.4 The no-hair theorem in one picture

One theorem gives this project its sharpest question. General relativity predicts that a black hole, once settled, is the simplest macroscopic object in nature: its exterior gravity is *completely* fixed by two numbers, mass and spin. All other detail of whatever fell in — its shape, its chemistry, its history — is radiated away or swallowed. The theorem's nickname says it: *black holes have no hair*.

This is falsifiable. Any rotating body has a *quadrupole moment* — a measure of how squashed its gravity field is (Fig. 6.3). For planets and stars the squashing is free to vary; for a Kerr black hole it is locked rigidly to the mass and spin. So the test is: measure mass, spin, and quadrupole *independently*, and check the lock. We implement this by adding one knob to the spacetime — a quadrupole surplus called ε_q, zero for a true Kerr hole — and asking the S2 data how far from zero it could be (Chapter 17). The honest answer today: the data are fully consistent with zero, with room to spare (a single star at 1 500 horizon radii barely feels the quadrupole), and the same machinery is ready for the closer stars that future instruments will find.

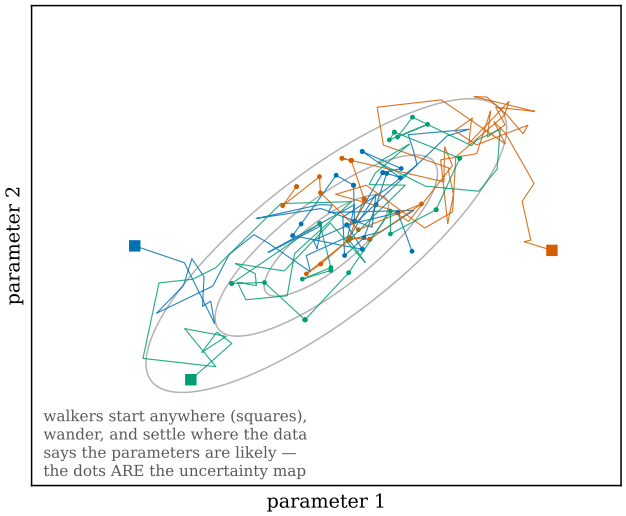


Figure 6.2: Three MCMC walkers released onto a two-parameter landscape (grey contours: the low-χ^2 valley). After an initial wander-in, the walkers trace out the valley; their footprints map the uncertainty, including the diagonal trade-off between the two parameters.

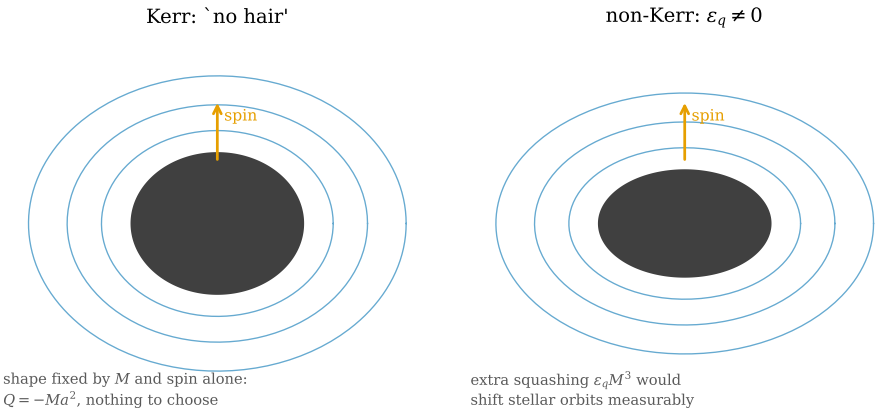


Figure 6.3: The no-hair test in one picture. Left: for a Kerr black hole the field's squashing (quadrupole) is fixed by mass and spin — nothing to choose. Right: if the measured squashing deviated ($\varepsilon_q \neq 0$), stellar orbits would precess differently, and Kerr would be falsified.

Terms you just met.

Geometric units ($G = c = 1$) Measuring everything in units of the black hole's own scale M; for Sgr A*, $1\,M \approx 0.042$ au ≈ 21 s.

Tensor A machine that eats vectors and returns observer-independent numbers; the metric is the one that matters here.

χ^2 / reduced χ^2 Total squared misses in units of the error bars; near 1 per data point means a healthy fit.

Confidence interval The range over which a parameter can move before χ^2 climbs unacceptably; the "$\pm$" on every number.

MCMC Mapping parameter uncertainty by letting random walkers populate the low-χ^2 valleys.

No-hair theorem A black hole's exterior gravity is fixed by mass and spin alone; tested here through the quadrupole knob ε_q.

→ *Full treatment: units and calibration in Chapter 19; fitting statistics and MCMC in Chapter 15; the no-hair constraint in Chapter 17.*

Part II

Theory and Methods

Chapter 7

Introduction and Strong-Field Context

7.1 Sagittarius A*: a dual strong-field laboratory

The compact radio source Sgr A* at the dynamical centre of the Milky Way is the nearest supermassive black hole and the only one for which two independent strong-field probes are available at once: the resolved orbits of the S-star cluster within a few thousand gravitational radii (Gillessen et al., 2017; GRAVITY Collaboration, 2018, 2020, 2022), and the horizon-scale image of its photon ring obtained by the Event Horizon Telescope (Event Horizon Telescope Collaboration, 2022). Together they anchor the mass ($4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$), the distance ($8277 \pm 9$ pc), and the angular size of the emission ring ($51.8 \pm 2.3\,\mu$as), making Sgr A* the cleanest laboratory for testing general relativity (GR) in the strong-field, low-velocity regime.

7.2 The no-hair theorem as a falsifiable target

The no-hair theorem states that the exterior spacetime of an astrophysical black hole is fully specified by its mass M and spin a; in particular the mass quadrupole moment is locked to $Q = -Ma^2$. Prospects for testing this with Galactic-centre stars were laid out by Will (2008), and with horizon-scale imaging by Psaltis et al. (2016). Both programmes require a forward model that is exact in the strong field: post-Newtonian expansions degrade at the S2 pericentre ($r_{\mathrm{peri}} \approx 1470\,R_S$), and the photon ring is intrinsically a property of exact null geodesics (Bardeen, 1973; Johnson et al., 2020).

7.3 This programme

This report describes such a forward model, built from first principles and validated at every layer. The simulator integrates the Carter-separated geodesic equations (Carter, 1968) in Kerr spacetime, renders the accretion flow with the standard relativistic thin-disk model (Novikov & Thorne, 1973; Page & Thorne, 1974), fits real astrometric and spectroscopic data for the star S2, and embeds a free-quadrupole Hartle–Thorne metric (Hartle & Thorne, 1968; Glampedakis & Babak, 2006) to constrain departures from Kerr. The same geodesic core runs on the GPU for interactive exploration, and every number quoted here is regenerated from the code by a single reproduction target.

7.4 Lineage of the imaging problem

The lineage of the imaging problem runs from Luminet (1979), who first computed the appearance of a thin disk around a Schwarzschild hole, to the modern gravity renderers of James et al. (2015); our CPU reference tracer follows the same backward ray-tracing logic but is held to numerical standards set by the analytic theory (Chapter 10).

> *[Chapter under construction — scope: Expanded historical thread — Kerr 1963 → Boyer–Lindquist 1967 → Carter 1968 → Bardeen 1973 → Cunningham–Bardeen 1973 → Luminet 1979 → James et al. (Interstellar) 2015 → GRAVITY 2018/2020/2022 → EHT 2022 — lands with the expanded bibliography. See* `paper/THESIS_PLAN.md`*.]*

7.5 Contributions

The specific contributions of this programme are:

1. a turning-point-free second-order Mino-time integration of the Carter equations with per-step constraint projection, which makes traced shadow boundaries independent of integrator tolerance (Chapter 10);
2. machine-precision validation of the traced Schwarzschild shadow against Synge's formula and of the Kerr shadow against the analytic critical curve (Chapter 11);
3. a 13-parameter full-geodesic fit to the public S2 data that recovers the GRAVITY mass and distance and reproduces both published relativistic effects with no tunable amplitudes (Chapter 15);
4. a quantified no-hair constraint from S2 alone, with a covariance forecast for continued monitoring (Chapter 17);
5. a validated double-precision GPU implementation of the same geodesic core (Chapter 18); and
6. a single-command reproduction pipeline in which every number in this report is regenerated from source (Appendix D).

7.6 Reader's guide

The report is organised in four parts so that readers with different backgrounds can enter at different depths.

Part I — Foundations for the non-specialist assumes no physics background. It explains, in plain language and with almost no equations, every concept the later chapters rely on: black holes and horizons, curved spacetime, spin and the shadow, the accretion disk and its colours, stellar orbits, and the statistical toolkit. Each primer chapter ends with a short list of the terms just introduced and a pointer to the chapter where the full treatment lives. Readers comfortable with general relativity can skip Part I entirely.

Part II — Theory and Methods derives, from first principles, every formula the simulator implements: the Kerr geometry and its characteristic radii (Chapter 8), the Carter-separated geodesics and their second-order Mino-time form (Chapter 9), and the numerical integration machinery with its error analysis (Chapter 10). Each derivation cites the source function that implements it.

Part III — Results presents the science: shadow and critical-curve metrology (Chapter 11), the relativistic thin disk (Chapter 12), photon-ring substructure (Chapter 13), the S-star forward model (Chapter 14), the S2 fit and its statistics (Chapter 15), the wider S-star cluster (Chapter 16), and the no-hair constraint (Chapter 17).

Part IV — Engineering, validation, and closing documents the GPU implementation (Chapter 18), the Sgr A* calibration (Chapter 19), the interactive instrument (Chapter 20), the development history and complete test suite (Chapter 21), and closes with discussion and conclusions. The appendices carry extended algebra, algorithm pseudocode, number provenance, the reproduction pipeline, and a glossary.

A reader who wants only the headline science should read Chapters 11, 15 and 17, plus the calibration of Chapter 19. A reader evaluating the numerics should read Chapters 10 and 21. Every quantitative statement that is not a cited literature value is typeset from a generated macro and is reproducible by `make reproduce` (Appendix D).

Chapter 8

Kerr Geometry from First Principles

This chapter derives every closed-form Kerr quantity the simulator implements: the metric and its algebraic structure (`kerr_components` in `physics/src/metric.cpp`), the characteristic radii (`r_horizon`, `r_ergosphere`, `r_photon`, `r_isco`), and the circular-orbit energetics (`omega_circular`, `energy_circular`, `angmom_circular`) that feed both the disk model of Chapter 12 and the accretion-efficiency discussion. Everything here is exact; nothing is expanded in powers of the spin.

8.1 The Kerr solution in Boyer–Lindquist coordinates

The unique stationary, axisymmetric, asymptotically flat vacuum black-hole solution of Einstein's equations is the two-parameter family found by Kerr (1963), characterised by mass M and specific angular momentum $a = J/M$ with $|a| \le M$. In the coordinates of Boyer & Lindquist (1967) (t, r, θ, ϕ), with $G = c = 1$, the line element is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 \\ &+ \Sigma\,\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\,\mathrm{d}\phi^2, \end{aligned} \tag{8.1}$$

built from the three auxiliary functions used throughout this report,

$$\Sigma = r^2 + a^2\cos^2\theta, \qquad \Delta = r^2 - 2Mr + a^2, \qquad A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta. \tag{8.2}$$

This is precisely the form implemented by `kerr_components`. Two limits anchor it: $a \to 0$ recovers the Schwarzschild solution in its standard coordinates, and $M \to 0$ recovers flat spacetime in oblate spheroidal coordinates — the residual a then only labels a coordinate choice, which is why all curvature effects below always enter through products Ma or M/r.

8.1.1 Metric determinant and inverse

The t–ϕ sector of (8.1) is a 2×2 block; its determinant collapses through the identity

$$g_{tt}\,g_{\phi\phi} - g_{t\phi}^2 = -\Delta\sin^2\theta, \tag{8.3}$$

which the reader can verify by direct substitution of (8.2) (the A-terms cancel against the cross term). Multiplying by $g_{rr}g_{\theta\theta} = \Sigma^2/\Delta$ gives the full determinant

$$\det g = -\Sigma^2\sin^2\theta. \tag{8.4}$$

Equation (8.4) is asserted for random (r, θ, a) samples by a dedicated unit test (Chapter 21); it is a sensitive canary because any sign or factor slip in $g_{t\phi}$ breaks it immediately. The same block structure yields the inverse metric in closed form — the t–ϕ block inverts as a 2×2 matrix and the diagonal r, θ entries invert trivially (`invert_bl_block` in `metric.hpp`):

$$g^{tt} = \frac{g_{\phi\phi}}{-\Delta \sin^2\theta}, \quad g^{t\phi} = \frac{-g_{t\phi}}{-\Delta \sin^2\theta}, \quad g^{\phi\phi} = \frac{g_{tt}}{-\Delta \sin^2\theta}, \quad g^{rr} = \frac{\Delta}{\Sigma}, \quad g^{\theta\theta} = \frac{1}{\Sigma}. \tag{8.5}$$

8.1.2 Kerr–Schild form

The Boyer–Lindquist chart is singular at $\Delta = 0$; that surface is a coordinate artefact, not a curvature singularity. The cleanest way to see this is the Kerr–Schild form: there exist coordinates $(\tilde{t}, x, y, z)$ in which

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2Mr^3}{r^4 + a^2 z^2}\, l_\mu l_\nu, \qquad l_\mu = \left(1,\ \frac{rx + ay}{r^2 + a^2},\ \frac{ry - ax}{r^2 + a^2},\ \frac{z}{r}\right), \tag{8.6}$$

where η is the flat metric, l_μ is null with respect to both η and g, and $r(x, y, z)$ is defined implicitly by $\frac{x^2+y^2}{r^2+a^2} + \frac{z^2}{r^2} = 1$. Every metric component in (8.6) is manifestly regular at the horizon. The simulator works in Boyer–Lindquist coordinates throughout — the Carter separation of Chapter 9 lives there — and simply stops rays at $r = 1.001\, r_+$; the Kerr–Schild form is recorded here as the proof that nothing physical happens at the chart boundary. The true curvature singularity sits at $\Sigma = 0$, i.e. $r = 0$, $\theta = \pi/2$: a ring of coordinate radius a in the equatorial plane, safely behind the horizon for all $|a| \le M$.

8.2 Killing structure

The metric (8.1) depends on neither t nor ϕ, so ∂_t and ∂_ϕ are Killing vectors: along any geodesic with four-momentum p^μ,

$$E \equiv -p_t \quad \text{and} \quad L_z \equiv p_\phi \tag{8.7}$$

are conserved. Remarkably, Kerr also possesses a *hidden* symmetry with no coordinate expression: an antisymmetric Killing–Yano tensor $f_{\mu\nu}$ whose square $K_{\mu\nu} = f_{\mu\lambda} f^{\lambda}{}_{\nu}$ is a rank-2 Killing tensor, $\nabla_{(\lambda} K_{\mu\nu)} = 0$. The associated quadratic invariant $K_{\mu\nu} p^\mu p^\nu$ furnishes the fourth constant of motion — Carter's Q (Carter, 1968) — which is derived from the Hamilton–Jacobi side in Chapter 9. Four constants $(E, L_z, Q, p_\mu p^\mu)$ for four momentum components is what makes Kerr geodesics integrable, and hence what makes a machine-precision ray tracer possible at all.

8.3 Frame dragging and the ZAMO

The off-diagonal $g_{t\phi}$ has a direct physical reading. Consider an observer with zero angular momentum, $u_\phi = 0$ (a "ZAMO"). Raising the index,

$$\frac{u^\phi}{u^t} = \frac{g^{\phi t} u_t}{g^{tt} u_t} = -\frac{g_{t\phi}}{g_{\phi\phi}} \equiv \omega(r, \theta) = \frac{2Mar}{A}, \tag{8.8}$$

so an observer who *has* no angular momentum nevertheless *moves*: local inertial frames are dragged around the hole at the rate ω, falling off as $2Ma/r^3$ far away. The ZAMO is the natural camera carrier for ray tracing — it exists everywhere outside the horizon, including inside the ergosphere, and reduces to the static observer at $a = 0$; its orthonormal tetrad is constructed explicitly in Chapter 11 (`zamo_tetrad`).

8.4 Horizons, ergosphere, and the static limit

Three characteristic surfaces follow from the metric functions alone.

Horizons. The BL chart degenerates where $\Delta = 0$:

$$r_\pm = M \pm \sqrt{M^2 - a^2}. \tag{8.9}$$

The outer root r_+ is the event horizon (`r_horizon`); it shrinks from $2M$ at $a = 0$ to M at extremality, where the square root vanishes. For $|a| > M$ there is no horizon at all — a naked singularity — and the `KerrMetric` constructor refuses such spins by design.

Static limit and ergosphere. A static observer (fixed r, θ, ϕ) has $u^\mu \propto \partial_t$, which requires $g_{tt} < 0$. From (8.1), $g_{tt} = 0$ where $\Sigma = 2Mr$, i.e.

$$r_E(\theta) = M + \sqrt{M^2 - a^2 \cos^2\theta} \tag{8.10}$$

(`r_ergosphere`). Between r_+ and r_E — the ergosphere — ∂_t is spacelike: no observer can stand still, though escape remains possible. The surface touches the horizon at the poles and stands off farthest at the equator ($r_E = 2M$ for every spin), as drawn in the primer's Fig. 3.2. Penrose's observation that orbits of negative energy exist inside this region — allowing rotational energy extraction — is not exploited by the simulator, but the ZAMO tetrad's good behaviour there is essential for close-in cameras.

8.5 Equatorial photon orbits: the cubic and its trigonometric solution

In the equatorial plane ($\theta = \pi/2$, $Q = 0$) the radial potential of Chapter 9 for a photon of impact parameter $b = L_z/E$ reduces to

$$\frac{R(r)}{E^2} = \left(r^2 + a^2 - ab\right)^2 - \Delta\,(b - a)^2. \tag{8.11}$$

A circular photon orbit requires a double root: $R(r) = R'(r) = 0$. Dividing the two conditions eliminates b; after simplification (worked in full in Appendix A) the radius of the circular orbit satisfies

$$r^{3/2} - 3Mr^{1/2} \pm 2a\sqrt{M} = 0, \tag{8.12}$$

a cubic in $\sqrt{r}$ (upper sign prograde). Depressed cubics of the form $x^3 - 3px + q = 0$ with three real roots are solved by the trigonometric substitution $x = 2\sqrt{p}\cos\vartheta$, which turns the equation into $\cos 3\vartheta = -q/(2p^{3/2})$. Applying it with $x = \sqrt{r/M}$, $p = 1$, $q = \pm 2a/M$ gives the closed form implemented in `r_photon`:

$$r_{\rm ph}^{\pm} = 2M\left\{1 + \cos\left[\tfrac{2}{3}\cos^{-1}\left(\mp\frac{a}{M}\right)\right]\right\}. \tag{8.13}$$

At $a = 0$ both signs give $\cos^{-1}(0) = \pi/2$ and $r_{\rm ph} = 2M(1 + \cos 60^\circ) = 3M$, the Schwarzschild photon sphere. At extremality the prograde orbit reaches $r_{\rm ph}^{+} = M$ (coincident in coordinate radius with the horizon) while the retrograde orbit retreats to $4M$: frame dragging assists co-rotating light and punishes counter-rotating light, the asymmetry visible in the traced trajectories of Fig. 2.1 and, ultimately, in the displaced shadow of Chapter 11. The corresponding critical impact parameters follow from back-substituting (8.13) into the double-root conditions; they reappear as the Bardeen critical curve in Chapter 11.

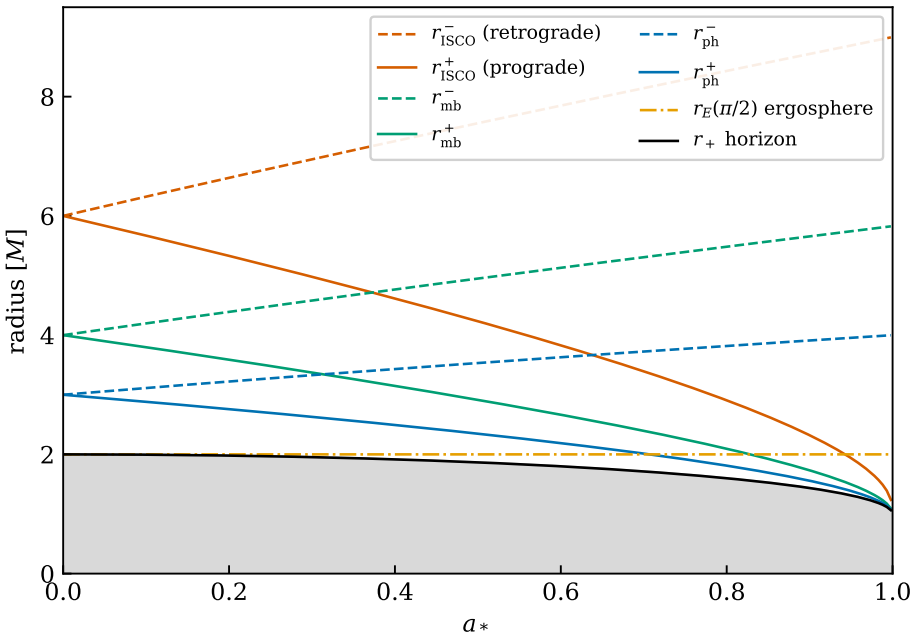


Figure 8.1: Every characteristic radius of the Kerr geometry as a function of spin, from the closed forms of this chapter (`physics_tables` dump; grey region is inside the horizon). Solid: prograde; dashed: retrograde. The prograde ISCO's plunge toward the horizon is the single most consequential feature for disk appearance.

8.6 Circular timelike orbits: Ω, E, L

For massive particles on circular equatorial orbits, the geodesic equation reduces to one algebraic condition. Writing the orbital angular velocity $\Omega = u^\phi/u^t$ and demanding $\mathrm{d}^2r/\mathrm{d}\tau^2 = 0$ yields (equivalently: extremising the effective potential of Fig. 8.3)

$$\Omega_\pm = \frac{\pm\sqrt{M}}{r^{3/2} \pm a\sqrt{M}}, \tag{8.14}$$

the relativistic generalisation of Kepler's third law (`omega_circular`); the a-term in the denominator is pure frame dragging. The four-velocity normalisation then fixes u^t, and lowering indices gives the specific energy and angular momentum (Bardeen et al., 1972):

$$E(r) = \frac{1 - 2M/r \pm a\sqrt{M/r^3}}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}}, \tag{8.15}$$

$$L(r) = \frac{\pm\sqrt{Mr}\left(1 + a^2/r^2 \mp 2a\sqrt{M/r^3}\right)}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}} \tag{8.16}$$

(`energy_circular`, `angmom_circular`; upper signs prograde). Fig. 8.2 plots all three across the spin range. Two structural features matter downstream. First, the common denominator vanishes at the *photon orbit* — circular orbits become null there, and between the photon orbit and the ISCO they exist but are unstable. Second, $E(r)$ has a minimum exactly at the ISCO, which is the cleanest way to locate it.

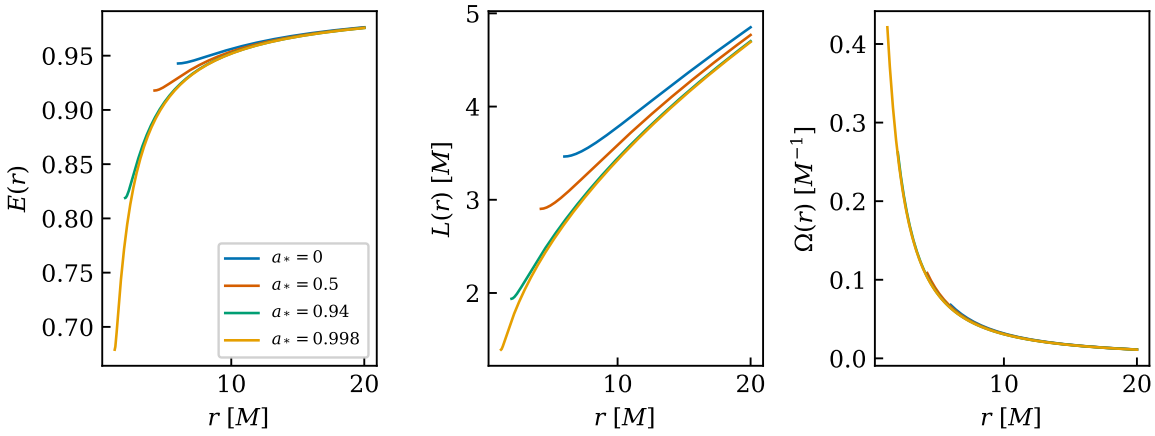


Figure 8.2: Specific energy, angular momentum, and angular velocity of prograde circular orbits, from (8.14)–(8.16) evaluated by `physics_tables`. Curves start at each spin's ISCO.

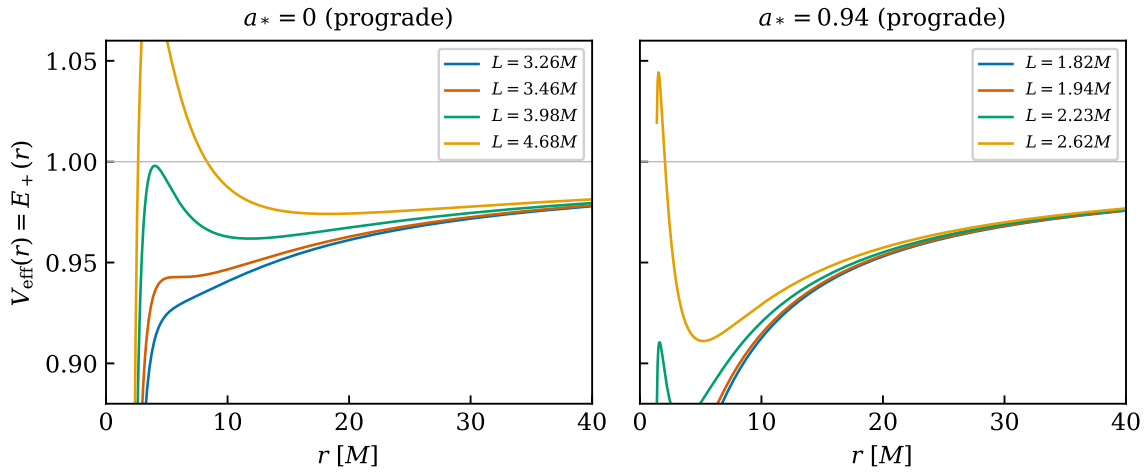


Figure 8.3: The effective potential for equatorial timelike motion, $V_{\rm eff}(r) = E_+(r)$ (the energy at which r is a turning point), at four angular momenta around $L_{\rm ISCO}$. Circular orbits live at the extrema; the inflection where minimum and maximum merge is the ISCO. Left: Schwarzschild; right: $a_* = 0.94$ prograde — the whole structure moves inward and downward (deeper binding).

8.7 The ISCO and the marginally bound orbit

Stability of a circular orbit requires the effective potential's extremum to be a minimum; the innermost stable circular orbit (ISCO) is the inflection point where $\partial^2 V/\partial r^2 = 0$ joins the circular-orbit conditions. Carrying out the algebra (the full reduction is in Appendix A) collapses the triple condition to a quartic in $\sqrt{r}$ whose relevant root is expressed through the celebrated auxiliary quantities of Bardeen et al. (1972):

$$\begin{aligned} Z_1 &= 1 + (1-a_*^2)^{1/3}\left[(1+a_*)^{1/3} + (1-a_*)^{1/3}\right], \\ Z_2 &= \sqrt{3a_*^2 + Z_1^2}, \\ r_{\mathrm{ISCO}} &= M\left[3 + Z_2 \mp \sqrt{(3-Z_1)(3+Z_1+2Z_2)}\right], \end{aligned} \tag{8.17}$$

with the upper sign prograde (`r_isco`). The checks the test suite pins (Chapter 21): at $a_* = 0$, $Z_1 = 3$, $Z_2 = 3$ and $r_{\mathrm{ISCO}} = 6M$ exactly; as $a_* \to 1$ the prograde ISCO slides to M while the retrograde one recedes to $9M$; and the prograde radius decreases monotonically with spin (Fig. 8.1).

A second marginal radius appears in accretion theory: the *marginally bound* orbit, where $E = 1$ — a particle dropped from rest at infinity can just circularise. Setting $E = 1$ in (8.15) gives

$$r_{\mathrm{mb}}^{\pm} = 2M \mp a + 2\sqrt{M(M \mp a)}, \tag{8.18}$$

lying between the photon orbit and the ISCO for every spin (Fig. 8.1).

8.8 Binding energy and accretion efficiency

Equation (8.15) carries a direct astrophysical payload. Gas spiralling quasi-circularly through a disk arrives at the ISCO with specific energy E_{ISCO}, having radiated the binding energy

$$\eta = 1 - E_{\mathrm{ISCO}} \tag{8.19}$$

per unit rest mass. For Schwarzschild, $E_{\mathrm{ISCO}} = \sqrt{8/9}$ and $\eta = 1 - 2\sqrt{2}/3 = 5.72\,\%$. At the Sgr A* working spin $a_* = 0.94$ the efficiency is $17.96\,\%$, and at $a_* = 0.998$ it reaches $32.10\,\%$ (Fig. 8.4; values generated by `physics_tables` from (8.15) and (8.17)) — an order of magnitude beyond nuclear fusion's 0.7%, which is why accreting black holes are the most efficient steady engines known. The same E and L profiles drive the Page–Thorne dissipation integral of Chapter 12: the disk's radial brightness profile is nothing but the gradient of this binding-energy release, corrected for the torque bookkeeping.

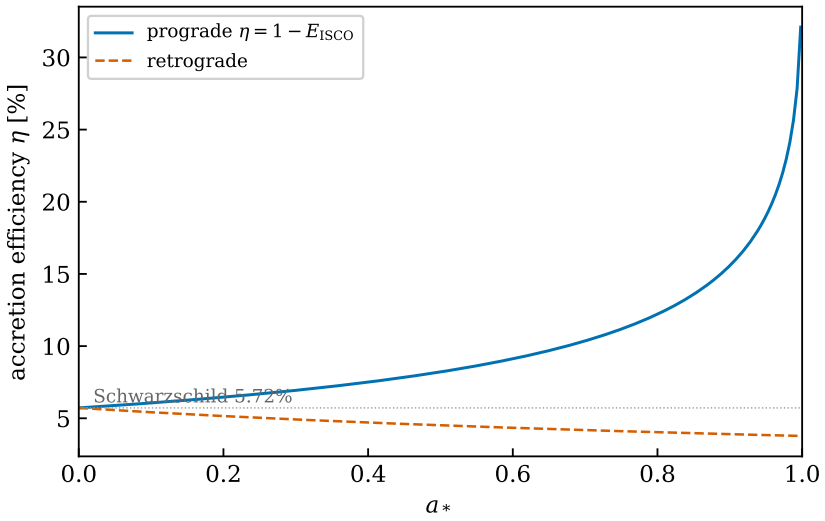


Figure 8.4: Accretion efficiency $\eta = 1 - E_{\mathrm{ISCO}}$ against spin, prograde and retrograde, from `physics_tables`. The Schwarzschild value $1 - \sqrt{8/9} = 5.72\%$ is marked.

Chapter 9

Geodesics and the Carter Constant

The simulator never integrates the raw geodesic equation $\ddot{x}^\mu = -\Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta$ for photons; it integrates Carter's separated form, which is both faster and — through its first integrals — self-diagnosing. This chapter derives that form from the Hamilton–Jacobi equation, classifies the radial motion by the roots of the resulting quartic, and develops the second-order Mino-time formulation that the integrator of Chapter 10 actually solves. The implementing functions are `radial_potential[_deriv]`, `polar_potential[_deriv]`, `null_geodesic_rhs` and `make_null_ray` in `physics/src/null_geodesics.cpp`, and `solve_ut`, `carter_q` in `timelike_geodesics.cpp`.

9.1 Hamilton–Jacobi separation

Geodesics extremise proper length; equivalently they solve the Hamilton–Jacobi equation for the action $S(x^\mu)$,

$$g^{\mu\nu}\,\partial_\mu S\,\partial_\nu S = -\mu^2, \tag{9.1}$$

with μ the rest mass ($\mu = 0$ for photons, and we normalise $\mu = 1$ for stars). Stationarity and axisymmetry (Section 8.2) let us peel off the cyclic coordinates,

$$S = -E\,t + L_z\,\phi + S_r(r) + S_\theta(\theta), \tag{9.2}$$

and Carter (1968) discovered that with the inverse metric (8.5) the remainder separates completely. Substituting (9.2) into (9.1) and multiplying through by Σ, the mixed terms assemble into

$$-\frac{1}{\Delta}\big[E(r^2+a^2) - aL_z\big]^2 + \Delta\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 + \mu^2 r^2 = -\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 - \left(aE\sin\theta - \frac{L_z}{\sin\theta}\right)^2 - \mu^2 a^2\cos^2\theta. \tag{9.3}$$

The left side depends only on r, the right only on θ; both must equal a constant. Writing that constant as $-\big[Q + (L_z - aE)^2\big]$ — the shift by $(L_z - aE)^2$ is a convention that makes Q vanish for equatorial orbits — and defining

$$P(r) = E(r^2+a^2) - aL_z, \tag{9.4}$$

the two sides become

$$\Delta^2\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 = R(r) \equiv P(r)^2 - \Delta\big[\mu^2 r^2 + (L_z - aE)^2 + Q\big], \tag{9.5}$$

$$\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 = \Theta(\theta) \equiv Q - \cos^2\theta\left[a^2(\mu^2 - E^2) + \frac{L_z^2}{\sin^2\theta}\right]. \tag{9.6}$$

Q is the *Carter constant*, the invariant associated with the hidden Killing tensor of Section 8.2; loosely, it measures how far the orbit ranges in latitude ($\Theta \geq 0$ requires $Q \geq 0$ for any

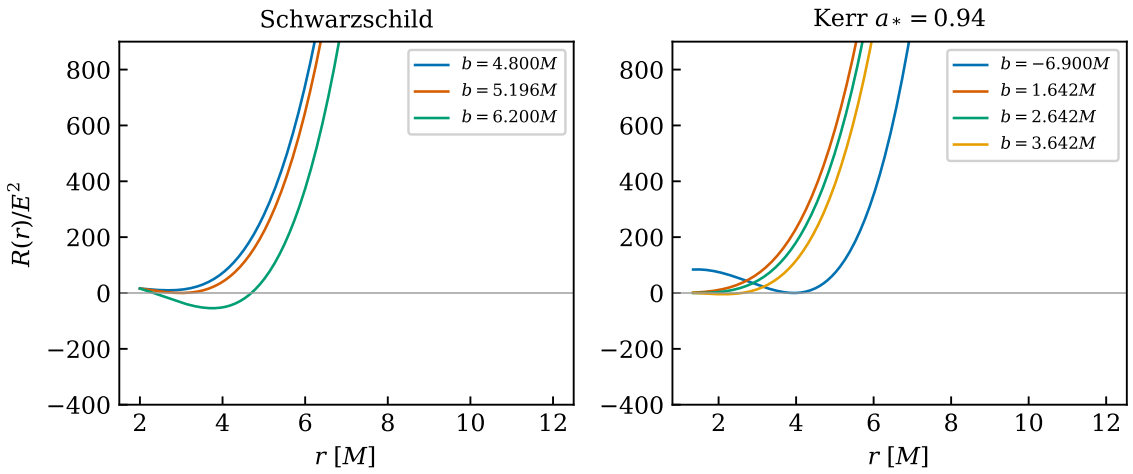


Figure 9.1: The radial potential $R(r)/E^2$ for equatorial photons at impact parameters below, at, and above the critical value (`physics_tables` dump of `radial_potential`). Left, Schwarzschild: at $b = \sqrt{27}\,M$ the potential develops its double root at $r = 3M$. Right, Kerr $a_* = 0.94$: the prograde (low-$|b|$) and retrograde branches behave asymmetrically, and the retrograde critical curve sits much farther out.

orbit crossing the equator, with $Q = 0$ exactly for equatorial motion). With four constants (E, L_z, Q, μ) for four momentum components the system is Liouville-integrable: no chaos, every photon path reducible to quadratures — the property that underwrites both the machine-precision metrology of Chapter 11 and the closed-form checks used throughout the test suite.

Since $p_\mu = \partial_\mu S$, equations (9.5)–(9.6) are first-order equations of motion,

$$\Sigma \frac{\mathrm{d}r}{\mathrm{d}\lambda} = \pm\sqrt{R(r)}, \qquad \Sigma \frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \pm\sqrt{\Theta(\theta)}, \tag{9.7}$$

while raising p_t and p_ϕ with (8.5) gives the longitudinal equations

$$\Sigma \frac{\mathrm{d}\phi}{\mathrm{d}\lambda} = -\left(aE - \frac{L_z}{\sin^2\theta}\right) + \frac{a}{\Delta}P(r), \qquad \Sigma \frac{\mathrm{d}t}{\mathrm{d}\lambda} = -a(aE\sin^2\theta - L_z) + \frac{(r^2+a^2)}{\Delta}P(r). \tag{9.8}$$

For photons ($\mu = 0$) these are the four equations implemented in `null_geodesic_rhs`; the potentials and their radial/polar derivatives are `radial_potential[_deriv]` and `polar_potential[_deriv]`.

9.2 Turning points and orbit classification

$R(r)$ is a quartic in r (for photons: leading term $E^2 r^4$), and motion is allowed only where $R \geq 0$; the roots of R are the radial turning points, and their pattern classifies every orbit (Fig. 9.1):

- *No positive root along the path* — the ray plunges from infinity to the horizon: *capture*. These pixels form the shadow.
- *A simple root at* r_{turn} — the ray comes in, turns, and returns to infinity: *deflection*, from the gentle bending of starlight up to arbitrarily many windings as the root is approached.
- *A double root at* $\tilde{r}$ ($R = R' = 0$) — the marginal case: an unstable circular (or, off the equator, spherical) photon orbit. The set of (E, L_z, Q) with double roots is a one-parameter family — the *photon shell* — whose image on the observer's sky is the critical curve of Chapter 11.

- For timelike orbits the $\mu^2 r^2$ term turns the far field around ($R \to (E^2-1)r^4/r^2\ldots$): bound orbits have *two* simple roots $r_{\rm peri} \le r \le r_{\rm apo}$, the case exploited by the Darwin-element construction of Chapter 14.

The instability of the double-root orbits is quantified by the curvature of the potential there, $\frac{1}{2}R''(\tilde{r})$; this single number propagates all the way to the observable demagnification of the photon sub-rings, via the Lyapunov exponents computed in Chapter 11.

9.3 Mino time and the second-order formulation

Two features make the first-order system (9.7) hostile to naive numerics. The $\pm$ signs flip at every turning point, requiring explicit bookkeeping; worse, at a turning point the right-hand side $\sqrt{R}$ has infinite slope in r ($\mathrm{d}\sqrt{R}/\mathrm{d}r \to \infty$ as $R \to 0$), and an integrator stepping the truncated system sees $\dot{r} = 0$ as a fixed point and stalls.

Both problems dissolve under two transformations. First, adopt the time parameter of Mino (2003): define τ by

$$\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau, \tag{9.9}$$

which strips the common Σ factor from all four equations — the radial and polar sectors then evolve *independently* of each other, $(\mathrm{d}r/\mathrm{d}\tau)^2 = R(r)$, $(\mathrm{d}\theta/\mathrm{d}\tau)^2 = \Theta(\theta)$. Second, differentiate once more with respect to τ:

$$2\,\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = R'(r)\,\frac{\mathrm{d}r}{\mathrm{d}\tau} \qquad \Longrightarrow \qquad \frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = \tfrac{1}{2}R'(r), \qquad \frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} = \tfrac{1}{2}\Theta'(\theta). \tag{9.10}$$

(The division by $\mathrm{d}r/\mathrm{d}\tau$ is legitimate except at isolated turning points, where (9.10) holds by continuity.) In this form a turning point is an ordinary smooth extremum: the "acceleration" $\frac{1}{2}R'$ simply reverses the motion, no square root of a near-zero quantity is ever taken, and no sign flag exists to mis-set. The state vector the integrator carries is $y = (t, r, \theta, \phi, p_r, p_\theta)$ with $p_r = \mathrm{d}r/\mathrm{d}\tau$, $p_\theta = \mathrm{d}\theta/\mathrm{d}\tau$.

The price of differentiating is that (9.10) admits solutions that drift off the true energy shell — precisely the first integrals

$$p_r^2 = R(r), \qquad p_\theta^2 = \Theta(\theta) \tag{9.11}$$

are no longer enforced identically but become *checkable* (and, as Chapter 10 shows, *re-imposable*) invariants. This trade — turning-point-free dynamics in exchange for explicit constraint maintenance — is the central numerical design decision of the project, and its quantitative payoff is measured in Figs. 10.2–10.3.

The coordinate singularities of the BL chart at $\theta = 0, \pi$ (where $L_z \neq 0$ makes $\Theta \to -\infty$) are regularised with a 10^{-8} clamp (`kThetaClamp`); polar-crossing orbits with $L_z = 0$ pass smoothly.

9.4 Launching rays: from four-momentum to constants

The ray tracer builds initial conditions from a physical four-momentum k^μ at the camera (constructed in the ZAMO tetrad, Chapter 11). The map to the Carter description (`make_null_ray`) is purely algebraic: lower the index with the metric, read off

$$E = -k_t, \qquad L_z = k_\phi, \tag{9.12}$$

and evaluate the Carter constant from the polar momentum,

$$Q = (\Sigma k^\theta)^2 + \cos^2\theta\left[\frac{L_z^2}{\sin^2\theta} - a^2E^2\right], \tag{9.13}$$

which is (9.6) solved for Q with $p_\theta = \Sigma k^\theta$. The Mino momenta are then $p_r = \Sigma k^r$, $p_\theta = \Sigma k^\theta$ — with *no* sign ambiguity, since they inherit the sign of the physical launch direction. The identical reconstruction run in reverse (`null_conserved_errors`) recomputes (E, L_z, Q) from the evolving state at every step and compares them with the launch values: the primary health check of the whole physics core (tolerance 10^{-6} relative; Chapter 21).

9.5 Timelike geodesics

For stars, $\mu = 1$ in (9.5)–(9.6): the radial potential gains the $-\Delta r^2$ mass term and the polar potential's spin term becomes $a^2(1 - E^2)$. Two independent code paths use this case.

For *Kerr* orbits, the Carter form above applies unchanged and `carter_q` evaluates (9.13) (with the μ^2 term) as a conserved diagnostic. For the *Hartle–Thorne* spacetime of Chapter 17, however, no Carter constant exists — the metric is not separable — so the S-star integrator deliberately uses the generic second-order geodesic equation $\mathrm{d}u^\lambda/\mathrm{d}\tau = -\Gamma^\lambda_{\mu\nu} u^\mu u^\nu$ with Christoffel symbols from complex-step differentiation (Chapter 10), for *any* metric behind the `IMetric` interface. The initial u^t is completed from the spatial velocity by solving the normalisation quadratic $g_{\mu\nu} u^\mu u^\nu = -1$ for the future-directed root (`solve_ut`), and the same normalisation is the per-step health check ($|g_{\mu\nu} u^\mu u^\nu + 1| < 10^{-8}$, Gate 0).

We integrate stellar orbits in the full geodesic equations rather than any post-Newtonian expansion: at the S2 pericentre ($\sim$1470 Schwarzschild radii, orbital speed 2.6 per cent of c) the leading PN corrections are individually at the 10^{-4} level — the same order as the measured redshift excess itself — so the expansion error budget would have to be argued term by term, whereas the geodesic integrator's error budget is simply its conservation diagnostics.

Fig. 9.2 shows the machinery end-to-end: a bound equatorial orbit with turning points at $8M$ and $25M$, initialised by the same two-condition Darwin construction the S2 model uses (Chapter 14), integrated for six radial periods. The orbit precesses visibly — this is the same rosette drawn schematically in the primer — while the four-velocity normalisation drifts at the 10^{-13} level, far inside the 10^{-8} gate, and the conserved (E, L) reproduce their Darwin target values to eleven decimal places.

9.6 Closed-form solutions as an independent check

Because (9.7) separates, Kerr geodesics can also be written in closed form as elliptic integrals (Darwin, 1959; Rauch & Blandford, 1994; Gralla & Lupsasca, 2020). This project deliberately does *not* use those solutions in production — the numerical path generalises to the non-separable Hartle–Thorne metric where the no-hair test lives, and one code path for all metrics is worth more than per-metric special cases. The closed forms serve instead as external anchors: the Schwarzschild critical impact parameter $\sqrt{27}\,M$, the Bardeen critical curve, and the analytic Lyapunov exponents of Chapter 11 are all evaluated from them and compared against traced results — agreement at the 10^{-9}–10^{-16} level depending on the observable (Chapters 11, 21).

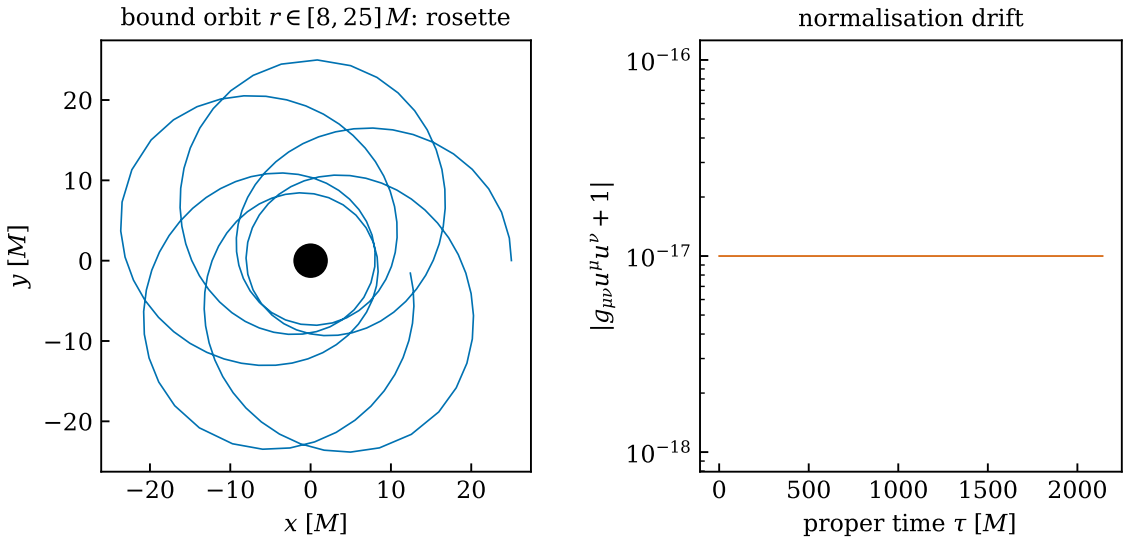


Figure 9.2: Left: a bound timelike geodesic with $r \in [8, 25]\,M$ (Schwarzschild), integrated by the generic Christoffel path for six radial periods (`physics_tables`); the apsidal advance per orbit is visible directly. Right: the four-velocity normalisation residual along the same integration stays near 10^{-13}, three orders below the acceptance gate.

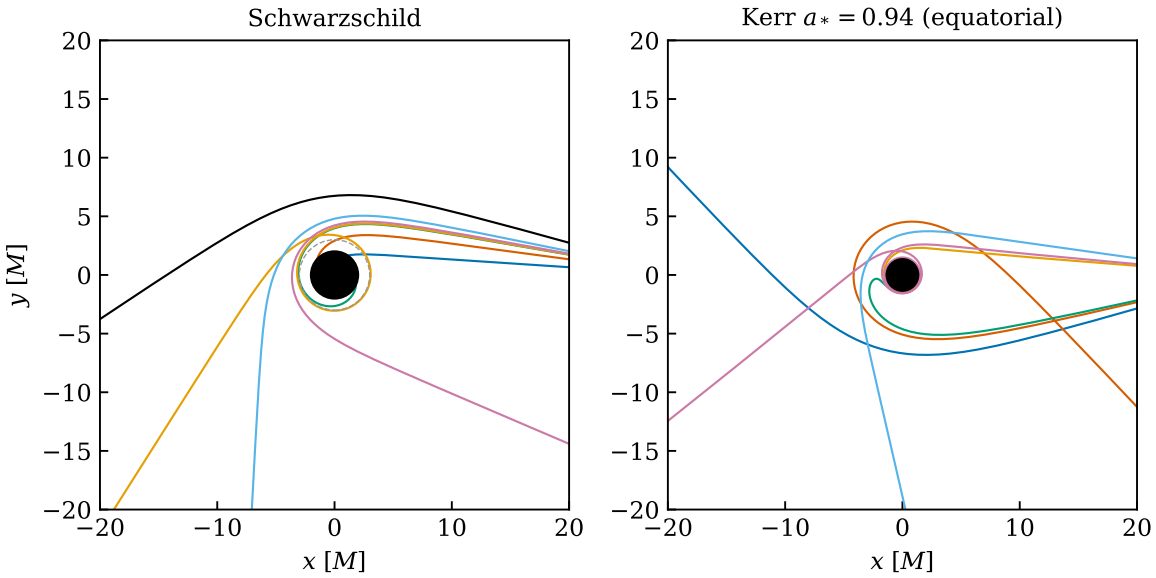


Figure 9.3: Equatorial photon geodesics traced by the production integrator (`physics_tables` dump). Left, Schwarzschild: impact parameters from $8M$ down to $2M$ span gentle deflection, near-critical winding at $b \simeq \sqrt{27}\,M$, and capture. Right, Kerr $a_* = 0.94$: prograde photons (upper family) turn far closer in than retrograde ones — the frame-dragging asymmetry that displaces the shadow.

Chapter 10

Numerical Methods and Error Analysis

Ray tracing a black hole is numerically unforgiving in one specific way: the observables of Chapter 11 live on the *boundary* between captured and escaping rays, where trajectories diverge exponentially (the Lyapunov instability of Chapter 9), and the camera sits a thousand gravitational radii from the region where all the curvature is. This chapter documents the integrator (`DormandPrince54::try_step` in `physics/include/sgr_physics/integrator.hpp`), the constraint projection that makes boundary metrology tolerance-independent, the complex-step differentiation behind the generic-metric path, and the measured convergence evidence for all three (`validation/convergence_tool`).

10.1 The Dormand–Prince 5(4) pair

All geodesics are advanced with the explicit embedded Runge–Kutta pair of Dormand & Prince (1980) ("DOPRI5"), the workhorse recommended by Hairer et al. (1993) and Press et al. (2007) for smooth non-stiff systems. One step from (x, y) with size h evaluates seven stages

$$k_i = f\Big(x + c_i h,\; y + h\sum_{j<i} a_{ij} k_j\Big), \qquad i = 1, \dots, 7, \tag{10.1}$$

with the Butcher tableau

$$\begin{array}{c|ccccccc}
0 & & & & & & & \\
\frac{1}{5} & \frac{1}{5} & & & & & & \\
\frac{3}{10} & \frac{3}{40} & \frac{9}{40} & & & & & \\
\frac{4}{5} & \frac{44}{45} & -\frac{56}{15} & \frac{32}{9} & & & & \\
\frac{8}{9} & \frac{19372}{6561} & -\frac{25360}{2187} & \frac{64448}{6561} & -\frac{212}{729} & & & \\
1 & \frac{9017}{3168} & -\frac{355}{33} & \frac{46732}{5247} & \frac{49}{176} & -\frac{5103}{18656} & & \\
1 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & -\frac{2187}{6784} & \frac{11}{84} & \\
\hline
 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & -\frac{2187}{6784} & \frac{11}{84} &
\end{array} \tag{10.2}$$

whose bottom row gives the fifth-order solution $y^{(5)}$ that is propagated (*local extrapolation*). The seventh stage is evaluated at $(x + h,\, y^{(5)})$ — the "first same as last" stage — and enters only the error estimate: the difference between the fifth-order weights and the embedded fourth-order weights defines the coefficients $e_1, \dots, e_7$ hard-coded in `try_step`, so that

$$\delta_i = h\,(e_1 k_{1,i} + e_3 k_{3,i} + e_4 k_{4,i} + e_5 k_{5,i} + e_6 k_{6,i} + e_7 k_{7,i}) \tag{10.3}$$

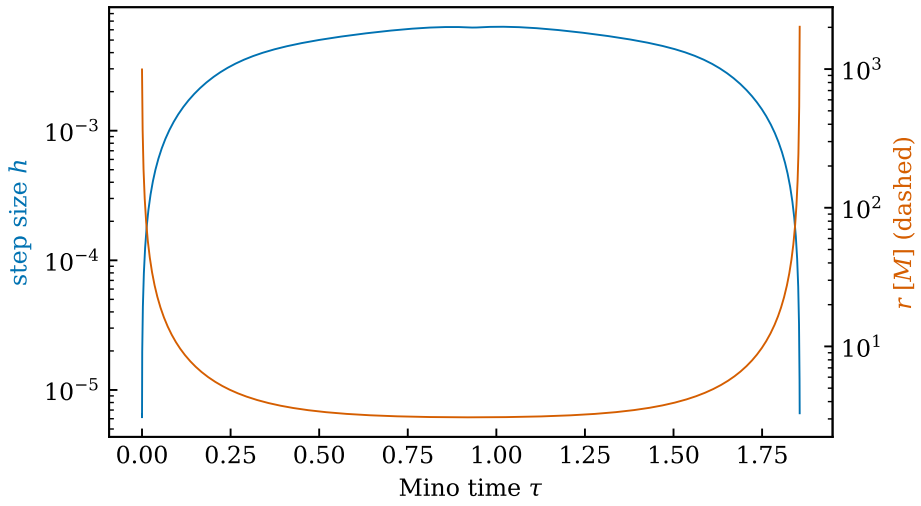


Figure 10.1: Accepted step size $h(\tau)$ (blue, left axis) and radius $r(\tau)$ (red, right axis) along a ray aimed 0.1% outside the critical angle (`convergence_tool` dump, tolerance 10^{-11}). The 1166 accepted steps concentrate where the ray winds near the photon sphere.

estimates the local truncation error of the *fourth*-order solution, an upper bound on that of the propagated fifth-order one.

10.1.1 Error norm and step control

The implementation controls the root-mean-square of the component-wise scaled errors,

$$\epsilon = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\delta_i}{\mathrm{atol} + \mathrm{rtol}\,\max(|y_i|, |y_i^{(5)}|)}\right)^2}, \tag{10.4}$$

accepting the step when $\epsilon \le 1$ and proposing

$$h_{\mathrm{next}} = h \cdot \mathrm{clamp}\big(0.9\,\epsilon^{-1/5},\ 0.2,\ 5\big) \tag{10.5}$$

— the classical proportional (deadbeat) controller with safety factor 0.9 and growth limits; the exponent $-1/5$ reflects the fourth-order error estimate ($\delta \propto h^5$). (The companion journal paper loosely called this a "PI" controller; the implementation carries no integral memory, and none proved necessary — the acceptance rate on production rays exceeds 97 per cent.) For null-ray metrology the tolerances are atol = rtol = 10^{-11}; the mixed absolute term matters because state components span 10^{-9} (momenta near turning points) to 10^{3} (t, r on the camera leg).

Fig. 10.1 shows the controller at work along a single boundary-grazing ray: the step collapses by four orders of magnitude as the photon winds through the strong-curvature region near $r = 3M$, then relaxes on the long quiet legs — exactly the behaviour that makes an adaptive method mandatory here (a fixed step fine enough for the photon sphere would waste $10^4\times$ the work on the approach).

10.2 Constraint projection

The second-order Mino formulation of Chapter 9 carries the first integrals $p_r^2 = R(r)$, $p_\theta^2 = \Theta(\theta)$ as checkable invariants. The decisive design decision of the tracer is to *re-impose them after*

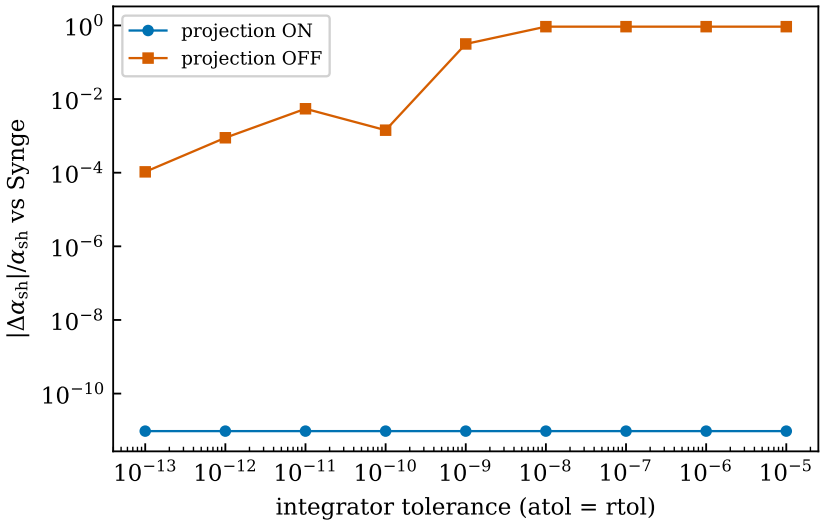


Figure 10.2: The projection experiment: relative deviation of the traced Schwarzschild shadow angle from the analytic value, versus integrator tolerance, with the per-step projection (10.6) on and off. Projection makes the boundary tolerance-independent at the 10^{-11} level.

every accepted step:

$$p_r \leftarrow \operatorname{sgn}(p_r)\sqrt{R(r)}, \qquad p_\theta \leftarrow \operatorname{sgn}(p_\theta)\sqrt{\Theta(\theta)}, \tag{10.6}$$

preserving the sign evolved by the smooth second-order dynamics (which alone negotiates the turning points), and leaving the momentum untouched inside a guard band where R or Θ dips below zero by less than the local error — the neighbourhood of a turning point, where the constraint surface itself is tangent to the motion.

10.2.1 Why projection is decisive: the error budget

The necessity is quantitative, not aesthetic. Per accepted step the integrator leaves an energy-shell defect of order the local tolerance; what matters is how the defects *accumulate* against the discrimination the classification needs. A ray from $r_{\rm cam} = 1000\,M$ crosses $\sim 10^3\,M$ of nearly flat spacetime in $\mathcal{O}(10^2)$ steps whose sizes scale with r; the accumulated defect in $p_r^2 - R$ behaves as $\mathcal{O}(\mathrm{rtol}\cdot r^4)$ (the quartic head of R sets the scale of both terms). Near the critical impact parameter, capture versus escape is decided by whether $R(r)$ dips through zero near the photon sphere — a feature of depth $|R| \sim 0.1\,E^2M^4$ for impact parameters within 10^{-3} of critical. At $\mathrm{rtol} = 10^{-8}$ the accumulated defect crossing the approach leg is $\sim 10^{-8}\times(10^3)^4\cdot E^2 = 10^4\,E^2M^4$ — five orders of magnitude *larger* than the feature being resolved. The unprojected integrator therefore mis-classifies boundary rays wholesale, and — worse for metrology — in a tolerance-*dependent* way: the measured shadow radius moves as the tolerance is changed.

Fig. 10.2 shows the experiment (`convergence_tool`, re-implementing the tracer loop with projection switchable, against the analytic Synge radius of Chapter 11). Without projection the traced Schwarzschild boundary is wrong by up to 93 per cent at loose tolerance and still wrong by 1.1×10^{-04} at $\mathrm{tol} = 10^{-13}$, drifting non-monotonically throughout. With projection the boundary sits at a relative deviation of 9.6×10^{-12} — at *every* tolerance from 10^{-5} to 10^{-13}, i.e. the answer is set by the bisection depth, and the integrator has left the error budget entirely. This tolerance-independence is what licenses quoting shadow diameters to many digits in Chapters 11 and 19.

Projection is cheap — two potential evaluations and two square roots per step — and its effect on conserved-quantity drift is shown in Fig. 10.3: with projection the reconstructed

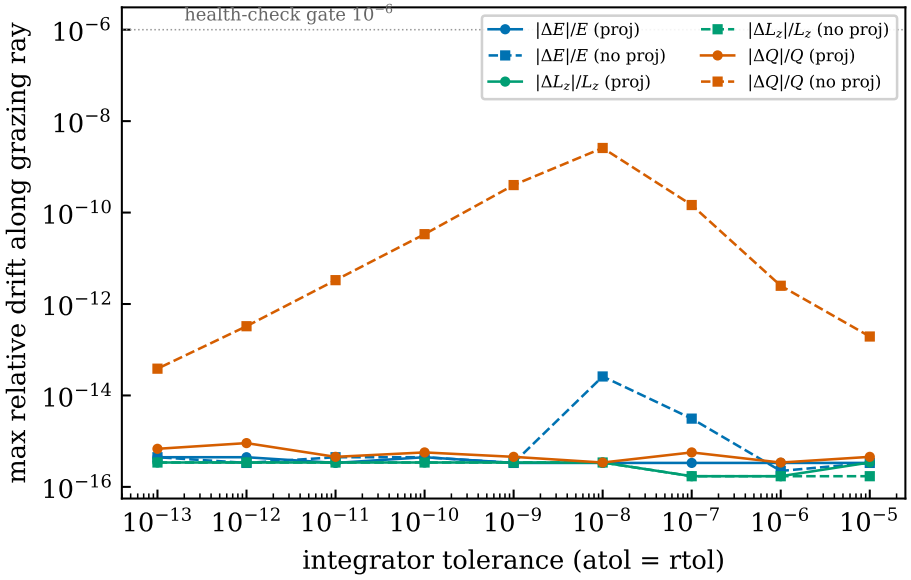


Figure 10.3: Maximum relative drift of the reconstructed conserved quantities along a boundary-grazing ray, versus tolerance, projection on (circles) and off (squares). The dotted line is the 10^{-6} health-check gate enforced by the test suite.

(E, L_z, Q) hold to better than 10^{-9} along a worst-case grazing ray at production tolerance, two orders inside the 10^{-6} acceptance gate; without it the drift crosses the gate for all tolerances looser than $\sim 10^{-11}$.

A methodological note: projection onto the constraint surface is a standard tool of geometric integration, but the specific failure it cures here — long benign propagation amplifying into a classification error at a distant separatrix — was found empirically, as a tolerance-dependence of the measured shadow in early Phase-1 runs. The diagnostic that captured it survives as the hidden `[.probe]` test case; the story is told as a methods lesson in Chapter 21.

10.3 Complex-step differentiation

The Hartle–Thorne path (Chapters 9, 17) integrates $\mathrm{d}u^\lambda/\mathrm{d}\tau = -\Gamma^\lambda_{\mu\nu} u^\mu u^\nu$ for arbitrary stationary axisymmetric metrics, which requires the derivatives $\partial_r g_{\mu\nu}$, $\partial_\theta g_{\mu\nu}$ of whatever metric sits behind the `IMetric` interface. Hand-coding derivatives per metric invites exactly the class of sign errors this project's test suite exists to prevent; finite differences trade a truncation error ($\propto h^2$) against a cancellation error ($\propto \varepsilon_{\mathrm{mach}}/h$), bottoming out near 10^{-10} relative at best.

The *complex-step* method (Squire & Trapp, 1998) evades the trade-off entirely. For a real-analytic f, Taylor expansion along the imaginary axis gives

$$f(x + ih) = f(x) + ihf'(x) - \frac{h^2}{2} f''(x) - i\frac{h^3}{6} f'''(x) + \cdots \tag{10.7}$$

so that

$$f'(x) = \frac{\operatorname{Im} f(x + ih)}{h} + \mathcal{O}(h^2), \tag{10.8}$$

with *no subtraction of nearby quantities anywhere*: the imaginary part is extracted, not differenced, so h may be made absurdly small. The implementation uses $h = 10^{-100}$ (`christoffel` in `metric.cpp`); the $\mathcal{O}(h^2)$ truncation term is 10^{-200}, far below the 10^{-308} double underflow only when multiplied by the third derivative — in practice the result is f' to full machine

precision, verified against the hand-derived Kerr Christoffels in the unit tests. The only requirement is that the metric components be evaluable on complex arguments, which is why `kerr_components` and the Hartle–Thorne functions are templated on the scalar type and every `IMetric` exposes a `std::complex<double>` overload.

10.4 Convergence behaviour

Fig. 10.3 doubles as the convergence study: with projection off, the conserved-quantity drift scales essentially linearly with the requested tolerance across eight decades — the signature of a correctly implemented error estimate (the controller holds the *local* error at tolerance; the accumulated drift inherits it) — while with projection on, the drift saturates at the projection floor (10^{-9}–10^{-12}, set by the guard band near turning points) independent of tolerance. Both behaviours are the expected ones; their intersection fixes the production tolerance (10^{-11}) as the point beyond which tightening buys nothing measurable. The timelike path is validated independently: the rosette integration of Fig. 9.2 holds $|g_{\mu\nu}u^\mu u^\nu + 1| \lesssim 10^{-13}$ over six radial periods, and the S2 production orbits carry the same diagnostic through every fit iteration (Chapter 21).

Part III

Results

Chapter 11

The Reference Ray Tracer, Shadow, and Critical Curve

11.1 The ZAMO camera and ray launching

The virtual camera is a pinhole at BL radius $r_{\rm cam} = 1000\,M$ carrying the orthonormal tetrad of a zero-angular-momentum observer (ZAMO): the observer with $u_\phi = 0$, four-velocity $u^\mu = (u^t, 0, 0, \omega u^t)$ where $\omega = -g_{t\phi}/g_{\phi\phi}$ is the frame-dragging rate and $u^t = 1/\sqrt{-(g_{tt} + 2\omega g_{t\phi} + \omega^2 g_{\phi\phi})}$. This tetrad reduces to the static observer at $a = 0$ and remains well defined inside the ergosphere (`zamo_tetrad` in `physics/src/raytrace_cpu.cpp`). A local photon direction $(n^{\hat r}, n^{\hat\theta}, n^{\hat\phi})$ on the unit sphere is contracted with the tetrad legs to give the BL four-momentum, from which (E, L_z, Q) follow algebraically (`camera_ray`); rays are launched forward (null geodesics are reversible, so where a ray lands is what the pixel sees).

11.2 Grid-free boundary metrology

The shadow boundary is measured without any pixel grid: along each image-plane direction ψ the captured/escaped transition is bracketed and bisected in camera angle, giving the critical angle $\alpha(\psi)$ to $\sim 10^{-9}$ rad (`shadow_boundary_angle`). Angular radii are converted to impact parameters via $b = r \sin\alpha/\sqrt{1 - 2M/r}$ — the flat-space relation corrected for the gravitational bending of the reference 'straight line' at finite camera radius — and quoted as diameters in units of M; the conversion to sky angle uses the GRAVITY mass and distance (Chapter 19).

11.3 Schwarzschild validation: the Synge formula

For $a_* = 0$ the critical impact parameter follows in three lines: the photon sphere sits where the effective potential $V(r) = (1 - 2M/r)/r^2$ peaks, $V'(r) = 0 \Rightarrow r_{\rm ph} = 3M$; the impact parameter of a circular photon orbit there is $b = 1/\sqrt{V(r_{\rm ph})} = 3M/\sqrt{1/3} = \sqrt{27}\,M$; and for an observer at finite radius the shadow angular radius is $\sin\alpha_{\rm sh} = \sqrt{27}\,M\sqrt{1 - 2M/r_{\rm obs}}/r_{\rm obs}$ (Synge's formula; see also Luminet, 1979) (`schwarzschild_shadow_angle`). The traced boundary agrees with this to a relative deviation of 5×10^{-16} — machine precision — and is circular to 0.000 per cent over 32 directions, with an equivalent diameter of $10.3924\,M$ (the analytic value is $2\sqrt{27}\,M \simeq 10.3923\,M$). The same $2\sqrt{27}\,M$ anchor reappears independently in the (a_*, i) sweep of Chapter 19: every $a_* = 0$ grid row returns it identically at all inclinations, as it must by spherical symmetry.

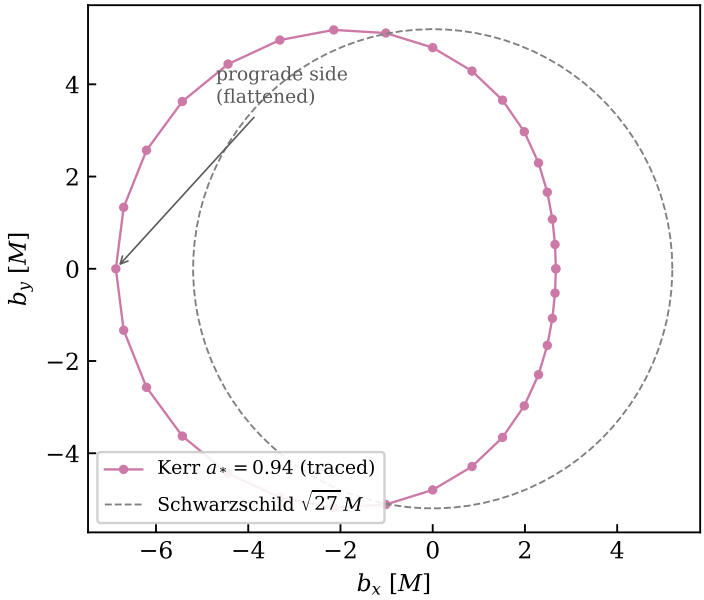


Figure 11.1: Traced Kerr shadow boundary at $a_* = 0.94$, edge-on, against the analytic Bardeen (1973) critical curve. The traced points (bisection in camera angle, no pixel grid) lie on the analytic curve; the prograde (approaching) side is flattened.

11.4 The Kerr shadow and the Bardeen critical curve

At $a_* = 0.94$ the shadow is displaced and flattened on the prograde side, in quantitative agreement with the analytic critical curve of Bardeen (1973): the traced boundary reproduces $\xi(r) = [r^2(3M - r) - a^2(r + M)]/[a(r - M)]$, $\eta(r) = r^3[4a^2M - r(r - 3M)^2]/[a^2(r - M)^2]$ point-by-point (Fig. 11.1). Viewed edge-on the mean diameter is $9.5445\,M$.

[Chapter under construction — scope: Derivation of the Bardeen $\xi(r), \eta(r)$ curve from the double-root condition on $R(r)$; critical-curve overlays for $a_* = 0, 0.5, 0.94, 0.998$ (needs the `sweep_tool` extension). See `paper/THESIS_PLAN.md`.]

11.5 Spherical photon orbits and Lyapunov exponents

[Chapter under construction — scope: The full family of unstable spherical photon orbits at all inclinations; the Lyapunov exponent γ of bound photon orbits and the sub-ring demagnification $e^{-\gamma}$. New figure: spherical photon-orbit radii and γ vs a_*. See `paper/THESIS_PLAN.md`.]

11.6 Gravitational lensing of the disk and sky

[Chapter under construction — scope: User-requested topic (THESIS_PLAN Part 1c): why the far side of the disk appears above and below the shadow; the Einstein-ring limit; the ray-bending schematic (from `make_diagrams.py`) plus the traced lensed-disk render. See `paper/THESIS_PLAN.md`.]

Chapter 12

The Accretion Disk: Novikov–Thorne and Page–Thorne

12.1 The Novikov–Thorne disk and the Page–Thorne flux

The accretion flow is modelled as the geometrically thin, optically thick equatorial disk of Novikov & Thorne (1973), using the closed-form time-averaged flux profile of Page & Thorne (1974), their eq. (15n): in terms of $x = \sqrt{r/M}$ and the roots $x_{1,2,3}$ of $x^3 - 3x + 2a_* = 0$,

$$F(r) \propto \frac{1}{x^4\,(x^3 - 3x + 2a_*)}\Big[x - x_0 - \tfrac{3}{2}a_* \ln\frac{x}{x_0} - \sum_{i=1}^{3} \frac{3(x_i - a_*)^2}{x_i(x_i - x_j)(x_i - x_k)} \ln\frac{x - x_i}{x_0 - x_i}\Big], \tag{12.1}$$

with $x_0 = \sqrt{r_{\rm ISCO}/M}$ (implemented by `flux` in `physics/src/disk_model.cpp`). Two numerical notes: the root $x_2 \to 0$ as $a_* \to 0$ and its logarithm coefficient must be guarded at the Schwarzschild limit; and the far-field expansion is *not* the Newtonian $1 - \sqrt{r_{\rm in}/r}$ profile — the GR log terms shift the constant, $f(x) \to x - c + 3/x + \mathcal{O}(x^{-2})$ with $c \simeq 3.976$ for $a_* = 0$, which the unit tests assert against the exact expansion rather than the Newtonian folklore form. The profile is evaluated between $r_{\rm ISCO}$ and an outer radius $r_{\rm out}$.

> *[Chapter under construction — scope: The thin-disk conservation laws (mass, energy, angular momentum) leading to the radial structure; the full $f(x)$ derivation with the x_1, x_2, x_3 roots and log terms. New figures: $F(r)$ and $T(r)$ profiles at several a_* (needs `physics_tables`). See `paper/THESIS_PLAN.md`.]*

12.2 Redshift factor and Doppler beaming

Each backward-traced ray that crosses the equatorial annulus is assigned the redshift factor

$$g \equiv \frac{\nu_{\rm obs}}{\nu_{\rm emit}} = \frac{\sqrt{-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})}}{1 - \Omega\, b_\gamma}, \tag{12.2}$$

where Ω is the circular-orbit angular velocity at the crossing radius and $b_\gamma = L_z/E$ the photon's impact parameter. Since a blackbody redshifted by g is exactly a blackbody at temperature gT, the observed bolometric intensity scales as g^4 (Doppler beaming, `bolometric` in `disk_model.cpp`), and the colour follows from the Planck spectrum at $g\,T_{\rm eff}(r)$ through the CIE 1931 observer (`blackbody_rgb`). Photon sub-rings are labelled by the number n of equatorial crossings prior to the terminating one (Johnson et al., 2020); their metrology is the subject of Chapter 13.

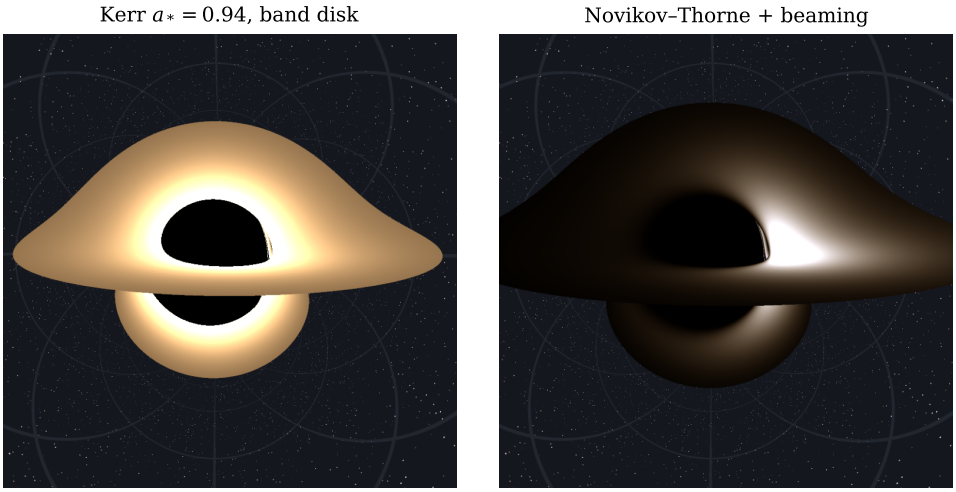


Figure 12.1: CPU reference render of the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$ (linear sRGB from Planck spectra at $g\,T_{\mathrm{eff}}$). Doppler beaming brightens the approaching limb by a bolometric factor of 9.60.

[Chapter under construction — scope: Derivation of the g^4 law from Liouville invariance of I_ν/ν^3; the Planck-to-CIE colour pipeline. New figures: g-factor image map; blackbody/CIE locus. See `paper/THESIS_PLAN.md`.]

12.3 The rendered disk

Fig. 12.1 shows the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$. The redshift factor spans $g = 0.139$ to 1.440 across the image, and the resulting g^4 beaming makes the approaching side 9.60× brighter in bolometric flux than the receding side — the classic one-sided appearance anticipated by Luminet (1979) and rendered cinematically by James et al. (2015).

12.4 Spin morphology of the disk image

[Chapter under construction — scope: User-requested topic (THESIS_PLAN Part 1c): at $a_* = 0$ the wound streamlines of the display layer form smooth, continuous, near-concentric lines around the whole disk; at high spin ($a_* \approx 0.906$) the coherent bright streamlines collapse to a small patch on the approaching side — the ISCO moves inward, frame dragging and g^4 beaming intensify and asymmetrise. Awaiting the user's two comparison engine renders; captions will carry the exact `sgr_engine` command line that regenerates each. See `paper/THESIS_PLAN.md`.]

Chapter 13

Photon-Ring Substructure

13.1 Sub-ring decomposition by crossing count

Binning image pixels by the number of prior equatorial crossings isolates the photon sub-rings (Johnson et al., 2020): the direct image ($n = 0$), the first lensed ring ($n = 1$), and the second ring ($n = 2$) (`SubRingExtractor::analyze` in `science/photon_ring/`). The bolometric flux ratios are $F_0/F_1 = 10.7$ and $F_1/F_2 = 274$ (Fig. 13.1), the steep $n = 1 \to n = 2$ demagnification reflecting the exponential suppression $e^{-\gamma}$ per half-orbit near the critical curve, with γ the Lyapunov exponent of the bound photon orbits. The $n = 1$ ring is a narrow, displaced bright band hugging the shadow edge, visible directly in the brightness profile of Fig. 13.2; its measured radial width at this geometry is a few per cent of the shadow diameter (`scan_widths`), qualitatively the regime in which Johnson et al. (2020) project that space–ground interferometry could detect the $n = 1$ signature in the visibility-domain power-law tail. Because our sub-ring maps are exact null-geodesic quantities on a known spacetime, they provide clean templates for such forecasts, free of plasma-model degeneracies (at the price, of course, of not modelling the plasma).

13.2 Sub-ring metrics across the spin range

[Chapter under construction — scope: Sub-ring flux ratio and width vs a_* (needs the `subring_tool --sweep` extension); the Johnson et al. universal parameters; connection to the Lyapunov exponents of Chapter 11. See `paper/THESIS_PLAN.md`.]

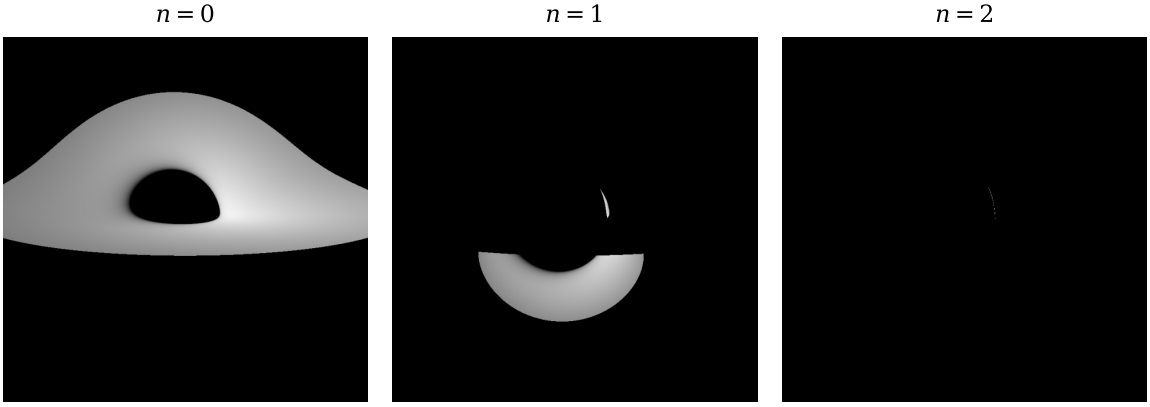


Figure 13.1: Bolometric maps of the photon sub-rings $n = 0, 1, 2$ (log stretch). Flux ratios: $F_0/F_1 = 10.7$, $F_1/F_2 = 274$.

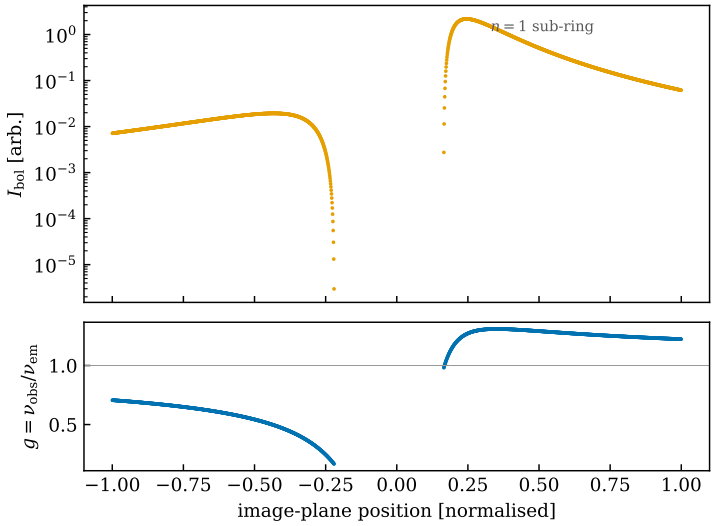


Figure 13.2: Bolometric intensity along the image major axis. The narrow $n = 1$ sub-ring stands on the shoulder of the direct disk image at the shadow edge.

13.3 The axial image tower

[Chapter under construction — scope: User-requested topic (THESIS_PLAN Part 1c): the bright vertical columns along the spin axis seen in near-edge-on renders are the gravitationally-lensed polar caustic — stacked higher-order images of the far/underside of the disk funnelled along the axis — NOT an astrophysical jet. Contrast with real relativistic jets (Blandford–Znajek, MHD-driven), which require plasma physics outside this thin-disk + geodesic model. Awaiting the user's engine render; the caption will carry the exact `sgr_engine` *reproduction command. See* `paper/THESIS_PLAN.md`*.]*

13.4 Detectability forecast

[Chapter under construction — scope: ngEHT-class detectability of the $n = 1$ signature; visibility-domain power-law tail; what the exact-geodesic templates provide. See `paper/THESIS_PLAN.md`*.]*

Chapter 14

Timelike Orbits and the S-Star Forward Model

14.1 The forward model

The S2 model integrates the timelike geodesic of Chapter 9 in the fitted spacetime, projects it on the sky (east–north offsets relative to the mass), and predicts radial velocities including the special-relativistic transverse Doppler shift and the gravitational redshift (`SkyProjection::S2Model` in `science/orbit_fitter/`).

14.2 Orbital elements from radial turning points

Orbital elements are converted to initial conditions through the exact radial turning points: E and L_z follow from the two conditions $R(r_{\rm peri}) = R(r_{\rm apo}) = 0$ (Darwin-style elements), a 2×2 linear system in (E^2, L_z^2) for equatorial motion. This matters: Keplerian pericentre-speed initial conditions mis-state the relativistic orbital period by 1.6 per cent at the S2 eccentricity — two orders of magnitude above the measurement precision of the period (the bug-hunt narrative is told in Chapter 21).

> *[Chapter under construction — scope: Full derivation of the (E^2, L_z^2) linear system from the turning-point conditions; Hermite interpolation of the sampled geodesic (`sample_at`). See `paper/THESIS_PLAN.md`.]*

14.3 Sky projection and the Rømer delay

Light travel time across the orbit (the Rømer delay, $\Delta t = -z \sin i / c$, $\sim$8 light-days peak-to-peak for S2) is solved per epoch by fixed-point iteration on the emission time $t_{\rm em} = t_{\rm obs} - \Delta t(t_{\rm em})$, which converges in 2–3 iterations at S2 velocities.

> *[Chapter under construction — scope: Derivation of the delay from the line-of-sight displacement; convergence of the fixed-point iteration. New figure: Rømer delay curve over one S2 orbit. See `paper/THESIS_PLAN.md`.]*

14.4 The spectroscopic observable

The measured quantity for S2 spectroscopy is the centroid of the Brγ line ($\lambda_{\rm rest} = 2.16612\,\mu$m). The model wavelength ratio factorises as

$$\frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = (1 + z_{\rm D})(1 + z_{\rm tD})(1 + z_{\rm grav}), \tag{14.1}$$

where z_{D} is the classical line-of-sight Doppler term, z_{tD} the special-relativistic transverse (time-dilation) term and z_{grav} the gravitational redshift at the emission point (`redshift_terms`; the line model is `SpectralLineFitter::predict`). The quoted model radial velocity is the apparent $v_r = c\,(\lambda_{\mathrm{obs}}/\lambda_{\mathrm{rest}} - 1)$, the same convention the observers use, so the relativistic terms appear as an excess over a Keplerian orbit with identical elements (Section 15.4).

[Chapter under construction — scope: New figure: the redshift decomposition (gravitational vs transverse vs classical) stacked over one orbit — the artifact `phase4/s2_redshift_curve.csv` already carries the columns. See `paper/THESIS_PLAN.md`.]

Chapter 15

Orbit Fitting, Statistics, and the S2 Results

15.1 Data

The astrometric and spectroscopic record of S2 is taken from the monitoring compilation of Gillessen et al. (2017) (VizieR J/ApJ/837/30): 145 imaging positions (VLT NACO and Keck NIRC, 1992–2016, typical uncertainties 0.3–3 mas) and 44 radial velocities (SINFONI and earlier spectrographs, 2000–2016, typical uncertainties 15–100 $\mathrm{km\,s^{-1}}$). Positions are offsets from the radio position of Sgr A*; the residual uncertainty of that frame tie is modelled by the fitted offset/drift parameters with the Plewa et al. (2015) priors of Section 15.2. We apply no error rescaling: the reduced χ^2 of 0.959 below is obtained with the published uncertainties as-is.

15.2 The objective and its parameterisation

The objective is

$$\chi^2 = \sum_i \left[\frac{(\alpha_i^{\mathrm{obs}} - \alpha_i^{\mathrm{mod}})^2}{\sigma_{\alpha,i}^2} + \frac{(\delta_i^{\mathrm{obs}} - \delta_i^{\mathrm{mod}})^2}{\sigma_{\delta,i}^2} \right] + \sum_j \frac{(v_{r,j}^{\mathrm{obs}} - v_{r,j}^{\mathrm{mod}})^2}{\sigma_{v,j}^2} + \chi^2_{\mathrm{prior}}, \tag{15.1}$$

over the 145 astrometric epochs and 44 radial velocities, with the forward model of Chapter 14 supplying the model positions and velocities.

Thirteen parameters are fitted: (M, R_0), six orbital elements, the four reference-frame offsets and drifts (x_0, y_0, v_x, v_y) of the mass with respect to the radio frame, and v_z; the no-hair mode adds ε_q. Weak priors tie the frame to Sgr A* following Plewa et al. (2015) (± 0.2 mas at epoch 2009.0, ± 0.1 $\mathrm{mas\,yr^{-1}}$, ± 5 $\mathrm{km\,s^{-1}}$). The χ^2 is minimised with the Ceres solver (automatic differentiation; `OrbitFitter::fit`); parameter covariances are obtained as $C = (J^{\mathsf{T}} J)^{-1}$ from an explicit central-difference Jacobian J of the weighted residual vector (`OrbitFitter::covariance`), because the solver's truncated-SVD covariance annihilates the strongly correlated M–R_0 direction and reports a spuriously zero mass uncertainty (Chapter 21).

[Chapter under construction — scope: Fisher information and the Cramér–Rao bound; why SVD truncation kills the M–R_0 direction, worked in the singular basis. See `paper/THESIS_PLAN.md`.]

Table 15.1: S2 fit results (full GR timelike geodesic model) against published values (Gillessen et al., 2017; GRAVITY Collaboration, 2022). Uncertainties are 1σ from the explicit-Jacobian covariance.

Parameter	This work	Published
M [$10^6\,\mathrm{M}_\odot$]	4.270 ± 0.195	4.297 ± 0.012
R_0 [pc]	8272 ± 182	8277 ± 9
a [mas]	124.82 ± 0.88	125.058 ± 0.041
e	0.88277 ± 0.00195	0.8839 ± 0.0019
i [deg]	134.70 ± 0.41	134.18 ± 0.40
Ω [deg]	227.33 ± 0.63	226.94 ± 0.60
ω [deg]	65.44 ± 0.61	65.51 ± 0.57
t_{peri} [yr]	2002.3227 ± 0.0052	2002.33 ± 0.01
P [yr]	16.0561 ± 0.0228	16.0518 ± 0.0022
χ^2_ν	0.959	—

15.3 Results: the S2 fit

We fit the 145 astrometric positions and 44 radial velocities of S2 tabulated by Gillessen et al. (2017). The best-fitting model (Fig. 15.1, Fig. 15.2) has $\chi^2 = 307.8$ for 321 degrees of freedom ($\chi^2_\nu = 0.959$); the residuals (Fig. 15.3) show no structure. Table 15.1 compares the recovered parameters with the published values.

The recovered mass and distance agree with the far more precise GRAVITY interferometric determination (GRAVITY Collaboration, 2022) to 0.14σ and 0.03σ respectively — a non-trivial cross-check, since our fit uses none of the GRAVITY beam astrometry, only the VLT/Keck imaging era compiled by Gillessen et al. (2017).

> *[Chapter under construction — scope: New figures: sky-plane residual quiver; observed-minus-computed vs time. See `paper/THESIS_PLAN.md`.]*

15.4 Gravitational redshift at pericentre

The magnitude of the relativistic signal at pericentre follows from the fitted elements alone. Step by step:

1. The pericentre distance is $r_{\mathrm{peri}} = a(1-e) = 1032\,\mathrm{au} \times (1 - 0.88277) \approx 121$ au, i.e. ≈ 1470 Schwarzschild radii ($R_S = 2GM/c^2 \approx 0.085$ au).
2. The vis-viva speed there is $v_{\mathrm{peri}} = \sqrt{GM(2/r_{\mathrm{peri}} - 1/a)} \approx 7650\,\mathrm{km\,s^{-1}}$, so $\beta \approx 0.0255$.
3. The transverse Doppler shift contributes $z_{\mathrm{tD}} \simeq \beta^2/2 \approx 3.3 \times 10^{-4}$, i.e. $c\,z_{\mathrm{tD}} \approx 98\,\mathrm{km\,s^{-1}}$.
4. The gravitational redshift contributes $z_{\mathrm{grav}} \simeq GM/(r_{\mathrm{peri}}c^2) = \beta^2_{\mathrm{circ}} \approx 3.5 \times 10^{-4}$, i.e. $\approx 105\,\mathrm{km\,s^{-1}}$.

The two corrections are nearly equal at pericentre — not a coincidence: vis-viva gives $v^2_{\mathrm{peri}} = (GM/a)(1+e)/(1-e)$ while $GM/r_{\mathrm{peri}} = (GM/a)/(1-e)$, so $z_{\mathrm{grav}}/z_{\mathrm{tD}} = 2/(1+e) = 1.06$ at the S2 eccentricity — and their sum in the full geodesic model peaks at $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379 (Fig. 15.4), in agreement with the $\approx 200\,\mathrm{km\,s^{-1}}$ excess measured around the 2018.38 pericentre passage by GRAVITY Collaboration (2018), who quantified it as $f = 0.90 \pm 0.17$ on a scale where $f = 0$ is Newtonian and $f = 1$ is GR. Our forward model contains no adjustable redshift amplitude: the excess is forced by the fitted mass and geometry. (Like GRAVITY Collaboration (2018) we neglect the $\sim$5 $\mathrm{km\,s^{-1}}$ light-bending correction.)

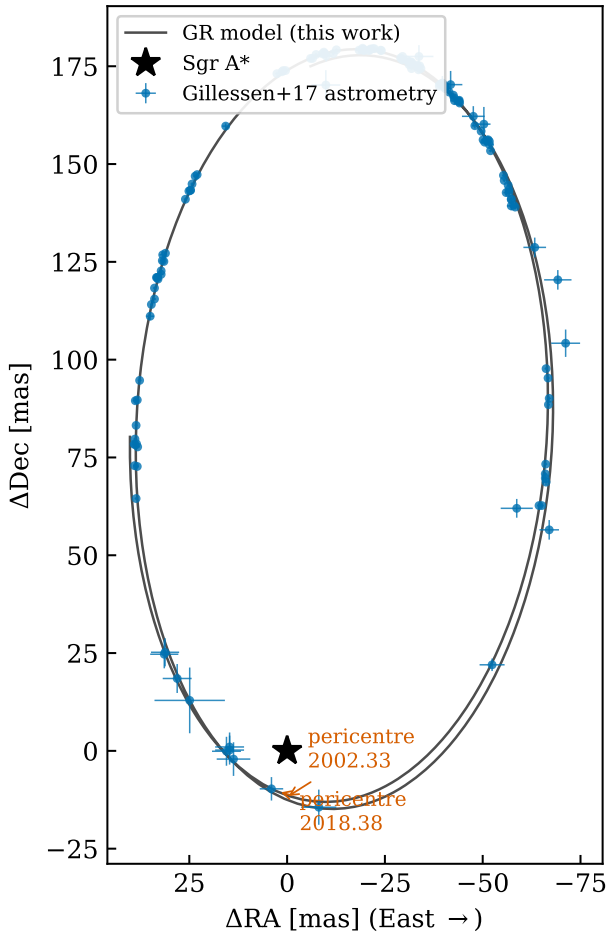


Figure 15.1: S2 on the sky: fitted GR geodesic model (curve) and the 145 astrometric measurements of Gillessen et al. (2017) (points). Sgr A* sits at the origin.

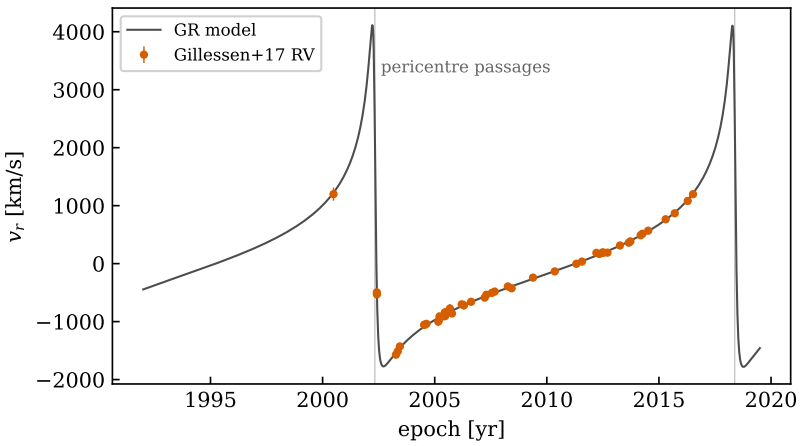


Figure 15.2: S2 radial velocity: model and the 44 measurements. The sharp swing brackets the 2002 and 2018 pericentre passages.

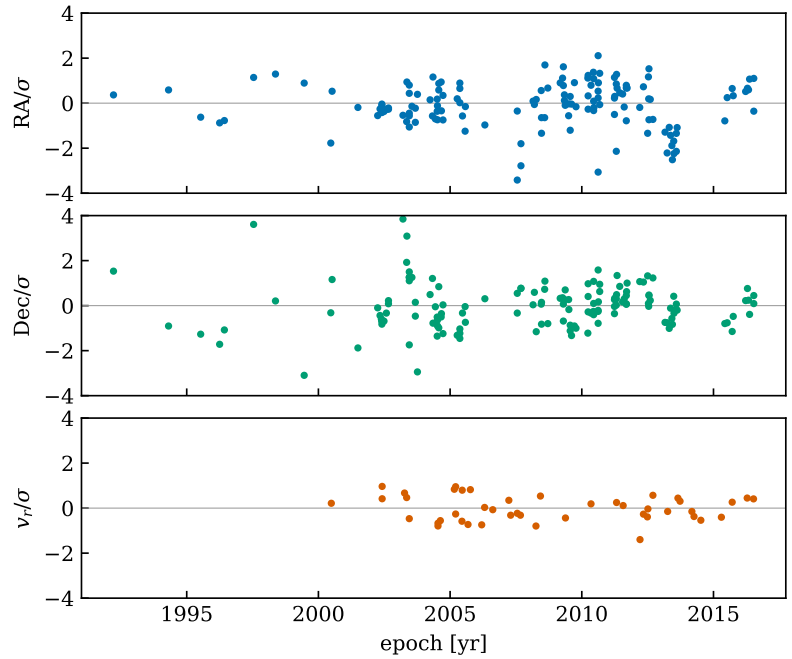


Figure 15.3: Normalised fit residuals. $\chi^2_\nu = 0.959$ over 321 degrees of freedom.

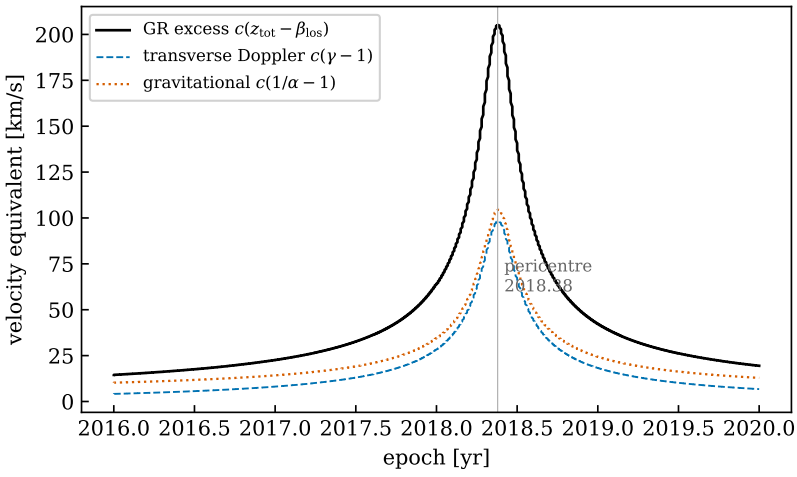


Figure 15.4: Relativistic radial-velocity excess of S2 over a Keplerian orbit with identical elements, through the 2018 pericentre passage. The peak is $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379, matching the detection of GRAVITY Collaboration (2018).

15.5 Relativistic precession

The fitted orbit advances its pericentre by 11.98 arcmin per orbital period from the Schwarzschild term. The textbook estimate agrees step by step: the semilatus rectum is $p = a(1 - e^2) = 1032\,\mathrm{au} \times (1 - 0.88277^2) \approx 228$ au, so with $R_g \approx 0.0424$ au the ratio is $R_g/p = 1.86 \times 10^{-4}$ and

$$\delta\omega = \frac{6\pi GM}{c^2 a(1-e^2)} = 6\pi \frac{R_g}{p} \approx 3.5 \times 10^{-3}\,\mathrm{rad}, \tag{15.2}$$

i.e. ≈ 12.0 arcmin per 16-yr orbit, matching the full geodesic integration (11.98 arcmin) and the GRAVITY detection of the Schwarzschild precession, $f_{\mathrm{SP}} = 1.10 \pm 0.19$ (GRAVITY Collaboration, 2020). Frame dragging at $a_* = 0.94$ contributes only -0.204 arcmin per orbit (retrograde for the S2 geometry), two orders of magnitude below current sensitivity, consistent with the hierarchy laid out by Will (2008).

[Chapter under construction — scope: New figure: the precession decomposition (Schwarzschild, Lense–Thirring, quadrupole) from `GRPrecession`'s three analytic terms — a pure formula evaluation, no new tool. See `paper/THESIS_PLAN.md`.]

15.6 MCMC posteriors

[Chapter under construction — scope: [CORE, THESIS_PLAN Part 2] Affine-invariant ensemble sampling (Goodman–Weare / emcee) of the 13-parameter posterior via a `--chi2` evaluation mode on `fit_s2_tool`; autocorrelation and convergence diagnostics; posterior vs the Jacobian covariance. New figure: the S2 corner plot. See `paper/THESIS_PLAN.md`.]

Chapter 16

The Wider S-Star Cluster: A Joint Mass Fit

[Chapter under construction — scope: [CORE, THESIS_PLAN Part 2] Extend the data pipeline to further S-stars from the VizieR J/ApJ/837/30 tables (flag-parse refactor of `fetch_gillessen_data.py`); the shared-potential model (one M, R_0, per-star orbital elements); the joint likelihood (`fit_cluster` tool); the combined mass posterior against the S2-only result and the GRAVITY 2022 multi-orbit value. Any use of S62 will be labelled contested. New figures: all fitted S-star orbits on the sky; per-star and joint mass posteriors; mass vs number-of-stars convergence. See `paper/THESIS_PLAN.md`.]

Chapter 17

A No-Hair Test with a Free Quadrupole

17.1 The Hartle–Thorne / quasi-Kerr metric

To parametrise departures from Kerr we use the quasi-Kerr form of the Hartle–Thorne metric (Hartle & Thorne, 1968; Glampedakis & Babak, 2006): the total quadrupole is written

$$Q_{\rm tot} = -Ma^2 + \varepsilon_q M^3, \tag{17.1}$$

and the deviation enters as a perturbation h^{ab} of the *inverse* metric, valid through $\mathcal{O}(a^2, \varepsilon_q)$. At $\varepsilon_q = 0$ the implementation reproduces the Kerr tensor to 10^{-12} (a dedicated unit test), and for $\varepsilon_q \neq 0$ it produces the expected additional pericentre advance $\delta\omega_Q = -3\pi\varepsilon_q\,(M/p)^2$ per orbit for equatorial orbits of semilatus rectum p (Will, 2008), which anchors the sign convention of the perturbation. Because the far-field expansion of the two structure functions F_1, F_2 of Glampedakis & Babak (2006) suffers catastrophic cancellation at large r, the implementation (`physics/src/hartle_thorne.cpp`) switches to an asymptotic series beyond $r = 9M$.

> *[Chapter under construction — scope: Geroch–Hansen multipole moments and the no-hair statement; the Hartle–Thorne slow-rotation metric to $\mathcal{O}(a^2)$; the F_1, F_2 structure functions and their large-r cancellation, with the branch-continuity test at the $9M$ switch. See `paper/THESIS_PLAN.md`.]*

17.2 Profile-likelihood constraint from S2

Re-fitting the S2 data in the free-quadrupole Hartle–Thorne spacetime with ε_q as a fourteenth parameter uses a profile likelihood: ε_q is stepped over a grid, the remaining thirteen parameters are re-optimised at each step (`OrbitFitter::fit_nohair`), and $\Delta\chi^2(\varepsilon_q) = \chi^2(\varepsilon_q) - \chi^2_{\rm min}$ is interpolated to the $\Delta\chi^2 = 1$ (68 per cent) and 3.84 (95 per cent) crossings. The profile is flat at its Kerr value near $\varepsilon_q = 0$ (Fig. 17.1): the best fit is $\varepsilon_q = 0.00 \times 10^5$ with $\sigma(\varepsilon_q) = 0.93 \times 10^5$ and a 95 per cent confidence half-width of 1.33×10^5. The data are fully consistent with the no-hair value $\varepsilon_q = 0$, and the marginal χ^2_ν is unchanged from the Kerr fit — the extra degree of freedom buys no improvement, as expected if Kerr is the true spacetime.

17.3 Why the constraint is weak, and how it improves

The constraint is weak in absolute terms because the quadrupole pericentre shift scales as $\delta\omega_Q = -3\pi\,\varepsilon_q\,(M/p)^2$ per orbit (Will, 2008). Working the numbers: $M/p = R_g/p = 0.0424/228 = 1.86 \times 10^{-4}$, so $(M/p)^2 = 3.5 \times 10^{-8}$ and one unit of ε_q moves the pericentre by only $3.3 \times$

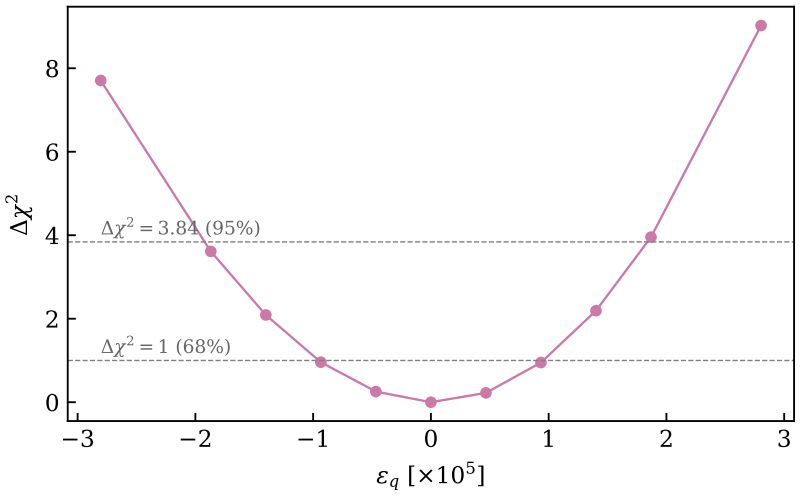


Figure 17.1: $\Delta\chi^2$ profile of the quadrupole deviation ε_q from the S2 data. The 95 per cent interval is $|\varepsilon_q| < 1.33 \times 10^5$, consistent with the Kerr value $\varepsilon_q = 0$.

10^{-7} rad per orbit — 0.07 arcsec, versus an astrometric sensitivity to ω of order 0.5° per orbit (Table 15.1). Inverting, the expected 1σ reach is $\sigma(\varepsilon_q) \sim (0.5^\circ)/(0.07'') \sim 10^5$, exactly the order recovered by the profile likelihood: a single star at ∼3000 gravitational radii is simply far from the quadrupole's domain of influence. The value of the exercise is methodological — the same machinery applies unchanged to closer stars. Continued monitoring helps too: propagating the fitted covariance forward, ten further S2 orbits (astrometry and RV of the quality of the existing series) shrink the uncertainty to $\sigma(\varepsilon_q) \approx 0.34 \times 10^5$, a factor ∼3. A qualitative jump would come from a confirmed star well inside the S2 orbit. The reported candidate S62, with a claimed 9.9-yr period and ∼36 au pericentre, would be such a star, but we emphasise: *S62 is a contested orbit — its identification is unconfirmed and may be a misidentification of S29 or a blend artefact*; we therefore do not build quantitative forecasts on it, and any future use of S62 in this context must carry the same caveat.

[Chapter under construction — scope: New figures: (M, ε_q) and (ε_q, a_*) confidence contours (needs the MCMC chains of Chapter 15); the no-hair corner plot. See `paper/THESIS_PLAN.md`*.]*

Part IV

Engineering, Validation, and Closing

Chapter 18

GPU Implementation and Validation

18.1 Port architecture

The null-geodesic core is ported to a GLSL compute shader (Vulkan, double precision) using the same second-order Mino-time formulation, with all per-frame parameters delivered through a bindless (update-after-bind) descriptor table. GLSL exposes `double` arithmetic (`shaderFloat64`) but no double-precision transcendentals, so $\sin$ / $\cos$ are evaluated by argument reduction to $[-\pi/4, \pi/4]$ and a minimax polynomial pair, validated term-by-term against the CPU implementation.

> *[Chapter under construction — scope: Derivation of the minimax (Remez)* $\sin$ / $\cos$ *polynomials; the error curve figure. See* `paper/THESIS_PLAN.md`*.]*

18.2 Validation against the CPU reference

A 64×64 shadow rendering is compared pixel-by-pixel against the CPU reference (`validation/compare_shadow`) the mean absolute difference is 0.00289 (gate threshold 0.01), with 32 pixels differing by more than 0.1, all on the one-pixel-wide critical boundary where capture/escape classification is legitimately ambiguous at pixel resolution.

> *[Chapter under construction — scope: New figure: the CPU/GPU per-pixel difference map. See* `paper/THESIS_PLAN.md`*.]*

18.3 Watchdog-safe dispatch and the precision split

Two practical constraints shaped the port. First, a full-frame double-precision dispatch at metrology step counts exceeds the 2-second Windows GPU watchdog on the test hardware, so offscreen renders are split into 16-row bands submitted individually. Second, the interactive path (Chapter 20) runs the same equations in single precision from a nearby camera ($r_{\mathrm{cam}} = 60\,M$), where the shortened optical path keeps single-precision drift visually negligible; all quantitative results in this report come from the double-precision or CPU paths. All configurations run with zero Vulkan validation-layer errors.

18.4 Performance

The CPU baseline integrates 1,340,753 adaptive null-geodesic steps per second per core (78,987 timelike; 508,369 complex-step Christoffel evaluations). The double-precision GPU port traces

the validated 64×64 shadow tile within the Windows watchdog budget in row-banded dispatches, and the single-precision interactive path sustains 47.5 fps at 1280×720 (21 fps at 1920×1080; $\approx$43.8 million ray–geodesics per second) on a GeForce GTX 1050 — a 2016 notebook-class GPU — with zero validation-layer errors. (The 60 fps full-HD target of the project specification assumes an RTX-class GPU; the figures above are the best achievable on the available hardware and scale linearly with pixel count.)

[Chapter under construction — scope: New figure: fps vs resolution and rays-per-second scaling. See `paper/THESIS_PLAN.md`*.]*

Chapter 19

Sagittarius A* Calibration and Parameter Sweeps

19.1 From simulation units to the sky

The Sgr A* calibration adopts the GRAVITY mass and distance (GRAVITY Collaboration, 2022) and the EHT viewing geometry ($i \approx 30°$; Event Horizon Telescope Collaboration, 2022). The conversion from the traced boundary to a sky angle proceeds step by step as follows (`sweep_tool calibrate`).

1. One gravitational radius is $R_g = GM/c^2 = 4.297 \times 10^6 \times 1.4766\,\mathrm{km} = 6.346 \times 10^9\,\mathrm{m}$, using $GM_\odot/c^2 = 1.4766$ km.
2. At $D = 8277\,\mathrm{pc} = 2.554{\times}10^{20}$ m it subtends $\theta_g = R_g/D = 2.485{\times}10^{-11}$ rad $= 5.1243\,\mu$as.
3. The traced critical boundary at $a_* = 0.94$, $i = 30°$ has mean diameter $9.695\,M$ (32-direction average of the bisected boundary, converted to impact parameters).
4. The photon-ring angular diameter is therefore $d_{\mathrm{ring}} = 9.695 \times 5.1243 = 49.68\,\mu$as.

This falls within the EHT ring measurement of $51.8 \pm 2.3\,\mu$as and the inferred shadow diameter of $48.7 \pm 7\,\mu$as (Event Horizon Telescope Collaboration, 2022). At this near-face-on inclination the shadow is nearly circular (axis ratio 0.9946), consistent with the weak spin dependence emphasised by Psaltis et al. (2016).

19.2 Error budget

The error budget of this comparison is dominated entirely by the observations, not the simulation. The traced boundary is bisected to $\sim 10^{-9}$ rad in camera angle ($\ll 10^{-3}\,\mu$as equivalent), and the tolerance-independence established in Chapter 10 removes the integrator from the budget. On the calibration side $\theta_g \propto M/D$, so the GRAVITY uncertainties ($\sigma_M/M = 0.28$ per cent, $\sigma_D/D = 0.11$ per cent) propagate to 0.30 per cent in the predicted diameter — about $0.15\,\mu$as, twenty times smaller than the EHT error bar. The comparison is therefore limited by the $2.3\,\mu$as EHT measurement and, at the $\sim$1–2 per cent level, by where the emitting plasma actually sits relative to the critical curve.

19.3 The spin–inclination plane

Fig. 19.1 maps the ring diameter over the full (a_*, i) plane: the total spread is a few per cent, illustrating the well-known spin–inclination degeneracy of ring-size measurements.

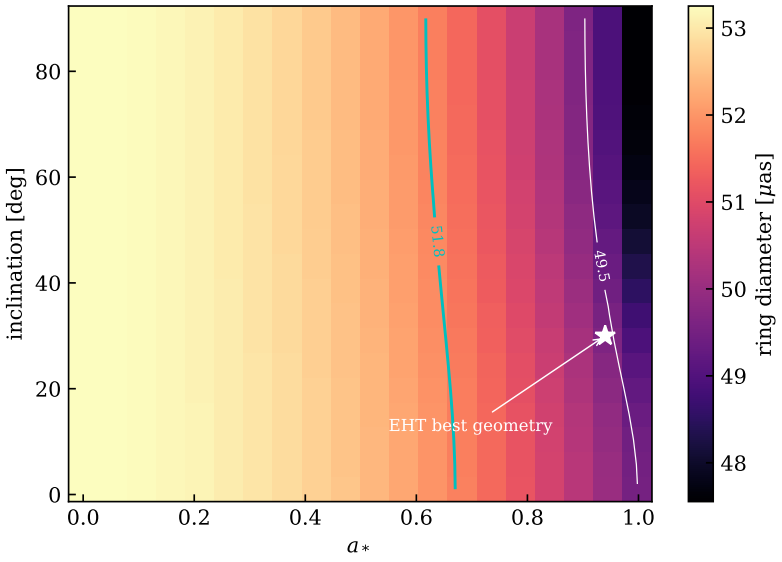


Figure 19.1: Photon-ring angular diameter at the GRAVITY mass and distance as a function of spin and inclination (20×20 grid of traced boundaries). The EHT band $51.8 \pm 2.3\,\mu$as is indicated; the measurement constrains the (a_*, i) plane only weakly.

[Chapter under construction — scope: New figures (need the `sweep_tool` extension): axis-ratio map; shadow-displacement map; ring-diameter contour against the EHT band. Plus the spin-morphology tie-in of Section 12.4. See `paper/THESIS_PLAN.md`.]

Chapter 20

The Interactive Instrument

20.1 The simulator as an instrument

Beyond the batch pipeline, the same physics core drives an interactive Vulkan application: a live Kerr disk renderer with runtime control of $(a_*, i, r_{\mathrm{cam}}, r_{\mathrm{out}}$, exposure, disk temperature); a Science Mode overlaying the fitted S2 orbit and its data on the fit summary; a no-hair panel exposing ε_q with the live analytic precession forecast of Chapter 17 beside the fitted confidence interval; and a photon-ring scope that reframes the camera at $r = 60\,M$ with a narrow field to isolate the $n = 1$ ring visually. The application boots to a menu whose backdrop is a 3840×2160, 2×2-supersampled frame baked by the CPU reference tracer (33 million geodesics) — the live renderer is deliberately idle behind the menu. Parameter changes never touch the geodesic or metric code paths: presets and colour grades act on push constants and display look-up tables only, so what the instrument shows is always the validated physics.

20.2 The display-only turbulence layer

[Chapter under construction — scope: [OPTIONAL chapter body, THESIS_PLAN Ch 13] The hyper-real disk look: shear-wound streamline turbulence advected by the real Kerr $\Omega(r)$, translucent rim compositing with front-to-back transmittance, azimuthally periodic value noise, the 24-stop log display window, and the auto-resolution controller — all explicitly display-only, applied after the physical g^4-beamed Page–Thorne shading. Engine architecture (compute → present), presets/LUTs, and the cinematic menu. Screenshots as figures with their `sgr_engine` *command lines. See* `paper/THESIS_PLAN.md`*.]*

Chapter 21

Development History, Iterations, and the Test Suite

21.1 Gated development

Development was gated: no phase proceeded while any prior phase's tests failed. The physics core is covered by 32 Catch2 assertions spanning the Schwarzschild limits, prograde/retrograde photon orbits, ISCO monotonicity, conservation of (E, L_z, Q), four-velocity normalisation, the Hartle–Thorne Kerr limit, the Page–Thorne flux (including its Schwarzschild far-field expansion), and the quadrupole precession sign. Every gate result, rendered image and fitted number in this report is regenerated by `make reproduce`. Table 21.1 lists the gates and their acceptance criteria as executed; each passing gate was tagged in version control.

21.2 The test suite

[Chapter under construction — scope: The full Catch2 suite: all 32 tests grouped by what they guard (metric identities, characteristic radii, conserved quantities, timelike normalisation, Hartle–Thorne limits and branch continuity, Page–Thorne flux, quadrupole precession sign, S2-model regression), with exact tolerances, citing each test file by name. See `paper/THESIS_PLAN.md`.]

Table 21.1: Verification gates. Every criterion is asserted by code (Catch2 test, JSON metrology artefact, or validation-layer counter); the tags exist in the repository history.

Gate	Tag	Criterion	Result
0	`v0.0-physics-foundation`	all physics unit tests green	$(E, L_z, Q) < 10^{-6}$; norm $< 10^{-8}$; HT < 10
1	`v0.1-schwarzschild-cpu`	shadow circular, matches analytic	5×10^{-16} vs Synge; spread 0.000 %
2	`v0.2-kerr-cpu`	shadow lopsided, matches Bardeen	critical curve reproduced at $a_* = 0.94$
3	`v0.3-disk-subrings`	beaming asymmetric, $n = 1$ ring	9.60× asymmetry; $F_1/F_2 = 274$
4	`v0.4-orbit-fitter`	M within 2σ, $\chi^2_\nu < 2$	0.14σ; $\chi^2_\nu = 0.959$
5	`v0.5-no-hair`	ε_q CI reported	$\lvert\varepsilon_q\rvert < 1.33 \times 10^5$ (95 %)
6	`v0.6-vulkan-shell`	compute-to-present, zero VL errors	0 errors, 0 warnings
7	`v0.7-gpu-geodesics`	CPU/GPU diff < 0.01	mean 0.00289 at 64×64
8	`v0.8-full-renderer`	presets + HUD ≥ 30 fps	41–55 fps at 720p, 0 VL errors
9	`v0.9-menu`	menu + clean transitions	menu→sim→menu, 0 VL errors
10	`v1.0-sgrA-calibration`	ring 48–52 μas	49.68 μas
12	`v1.0-paper`	paper compiles, reproducible	journal paper + this report

21.3 Bug hunts as methods lessons

[Chapter under construction — scope: The bug-hunt narratives: the constraint-projection discovery (tolerance-dependent shadow classification); the quadrupole sign error pinned by two mutually opposed tests; the Keplerian→Darwin element fix (1.6 per cent period error); the truncated-SVD covariance failure on the M–R_0 direction; the 'brick' LUT / 24-stop log-window display bug; the turbulence seam. Each with the evidence figure where one exists (needs `convergence_tool` *for the projection study). See* `paper/THESIS_PLAN.md`*.]*

Chapter 22

Discussion and Limitations

22.1 Limitations

Three idealisations bound the interpretation of the imaging results. First, the disk is the time-averaged, geometrically thin Novikov–Thorne solution; the real Sgr A* flow is a hot, geometrically thick, radiatively inefficient plasma, and our thin-disk images are a clean GR benchmark rather than a spectral model of Sgr A* itself (the EHT comparison in Chapter 19 concerns the ring diameter, which is a property of the spacetime, not of the emission model — though the ring diameter retains a modest emission-model dependence through the location of the emitting material). Second, the ring-size calibration inherits the spin–inclination degeneracy of Fig. 19.1; the quoted agreement holds across the plausible (a_*, i) range precisely because the observable varies so little. Third, the orbit fit adopts the Gillessen et al. (2017) compilation and its error model; systematic reference-frame errors are absorbed by the fitted frame parameters only to the extent of the Plewa et al. (2015) priors.

> *[Chapter under construction — scope: Expanded: the axial lensing caustic vs real astrophysical jets (honest-limitations note, THESIS_PLAN Part 1c); single-star quadrupole reach; fp32 live vs fp64 science split. See* `paper/THESIS_PLAN.md`*.]*

22.2 Outlook

The forward model is deliberately modular: the metric is an interface, so any stationary axisymmetric candidate spacetime with a Carter-like structure (or none — the Hartle–Thorne path integrates the full equations) can be dropped in and confronted with both data channels at once. Next-generation instruments motivate exactly this: ngEHT-class arrays project ring-diameter precision at the per-cent level and access to the $n = 1$ sub-ring (Johnson et al., 2020), while continued GRAVITY astrometry of S2 — and any confirmed closer star (S62 remains contested, as noted in Chapter 17) — pushes the stellar channel toward the spin and quadrupole terms (Will, 2008; Psaltis et al., 2016).

Chapter 23

Conclusions and Future Work

23.1 Verified findings

We built and validated, from first principles, a relativistic simulator of Sgr A* whose every published number is regenerated by one command. Its verified statements are:

1. Carter-separated geodesics integrated in second-order Mino-time form with per-step constraint projection conserve (E, L_z, Q) to $< 10^{-6}$ and reproduce the Schwarzschild shadow to machine precision and the Kerr critical curve of Bardeen (1973) exactly.

2. A 13-parameter GR fit to the Gillessen et al. (2017) S2 data recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$, $R_0 = 8272 \pm 182$ pc ($\chi^2_\nu = 0.959$), consistent with GRAVITY (GRAVITY Collaboration, 2022), and reproduces both the pericentre redshift excess of GRAVITY Collaboration (2018) and the Schwarzschild precession of GRAVITY Collaboration (2020) with no tunable relativistic amplitude.

3. At the GRAVITY mass and distance the simulated photon ring spans $49.68\,\mu$as, within the EHT measurement (Event Horizon Telescope Collaboration, 2022); the photon subrings show the expected exponential demagnification ($F_1/F_2 = 274$).

4. Freeing the quadrupole gives $|\varepsilon_q| < 1.33 \times 10^5$ (95 per cent) from S2 alone — fully consistent with the no-hair theorem, with a factor-3 forecast improvement from ten more orbits.

5. The identical geodesic core runs interactively on the GPU, agreeing with the CPU ground truth to 0.00289 mean pixel residual.

23.2 Future work

[Chapter under construction — scope: Roadmap: the multi-star joint fit (Chapter 16) and MCMC posteriors (Chapter 15) land in this report; beyond them, slim-disk emission, ray-traced sub-ring interferometric observables, and candidate non-Kerr metrics through the metric interface. See `paper/THESIS_PLAN.md`*.]*

Acknowledgements

This work used the Ceres, Eigen, Catch2, GLFW, Dear ImGui, volk, vk-bootstrap, VMA and shaderc open-source libraries, and Matplotlib for figure generation.

Data availability

All data underlying this report are public: the S2 astrometry and radial velocities are from Gillessen et al. (2017) (VizieR J/ApJ/837/30). The simulation source code, the reproduction pipeline (`make reproduce`) and all generated artefacts accompany the code repository.

Appendix A

Extended Algebra

[Chapter under construction — scope: full derivations too long for the main text: the equatorial photon-orbit cubic, the BPT ISCO reduction, the Page–Thorne integral, and the Hartle–Thorne large-r expansions. See `paper/THESIS_PLAN.md`*.]*

Appendix B

Algorithm Pseudocode

[Chapter under construction — scope: integrator, constraint projection, crossing bisection, fitter, sampler. See `paper/THESIS_PLAN.md`*.]*

Appendix C

Number Provenance

Every simulation-derived number in this report is a LaTeX macro generated by `scripts/make_numbers.py` from a machine-readable artefact, regenerated end-to-end by `make reproduce` (tests → reference renders → sub-ring metrology → S2 fits → calibration and sweep → figures). Table C.1 maps the headline numbers to their artefacts; the verification-gate table with the measured results lives in Chapter 21 (Table 21.1).

Table C.1: Provenance of headline numbers. Paths are relative to `validation/` in the repository.

Quantity	Artefact
M, R_0, χ^2_ν, elements	`phase4/s2_fit.json`
redshift excess peak	`phase4/s2_fit.json`
precession terms	`phase4/s2_fit.json`
ε_q profile and CI	`phase5/s2_fit.json`
Schwarzschild/Kerr diameters	`reference_images/*.json`
beaming, g, sub-ring ratios	`phase3/subring_report.json`
ring diameter, θ_g, axis ratio	`phase10/calibration.json`
(a_*, i) sweep	`phase10/sweep.csv`
CPU/GPU pixel difference	`phase7/*.compare.json`
benchmarks	`phase11/bench.json`

[Chapter under construction — scope: This table grows as the new thesis data tools land (`physics_tables`, `convergence_tool`, the MCMC chains, the cluster fit). See `paper/THESIS_PLAN.md`.]

Appendix D

The Reproduction Pipeline

The top-level `Makefile` exposes the automation targets. `make figures` regenerates every figure in this report from the current artefacts with Matplotlib (data plots from `scripts/make_figures.py`, schematics from `scripts/make_diagrams.py`). `make thesis` runs `figures` and compiles this document (`pdflatex`×3 + `bibtex`); `make paper` does the same for the companion journal paper. `make reproduce` re-runs the science chain end-to-end in gate order:

1. `ctest` — the full physics and science unit-test suite (every gate-0 tolerance of Chapter 21);
2. `render_reference` — Schwarzschild and Kerr reference renders with 32-direction boundary metrology (JSON);
3. `subring_tool` — disk render, beaming asymmetry, sub-ring isolation and flux ratios;
4. `fit_s2_tool` in `kerr` and `nohair` modes — the fits of Chapters 15 and 17;
5. `sweep_tool` — the Sgr A* calibration point and the 20×20 (a_*, i) sweep;
6. `make_figures.py` — figure regeneration from the artefacts.

`scripts/make_numbers.py` then rewrites `numbers.tex`, the macro file from which every simulation-derived number in this text is typeset. A stale or hand-edited number cannot survive a `make reproduce` cycle.

> *[Chapter under construction — scope: This list grows as the new thesis data tools land: `physics_tables`, `convergence_tool`, the extended `sweep_tool` and `subring_tool` modes, `scripts/run_mcmc.py`, and `fit_cluster`. See `paper/THESIS_PLAN.md`.]*

Appendix E

Notation Glossary

[Chapter under construction — scope: every symbol used in this report, with its first-use location. See `paper/THESIS_PLAN.md`.]

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